Faculty of Science - Software Engineering Teaching Unit
University of Kelaniya, Sri Lanka

SENG 34213

System Development Project

Test Case Register - Gaming Castle

1. Introduction and Conventions

1.1 Test case ID scheme

IDs have the form TC-<SVC>-<TYPE>-<NNN>. SVC is GW, USR, BKG, PAY, TRN, LOY or NTF. TYPE is UNIT, INT (integration) or SEC (security filter tests). NNN counts in source order within a service and type. These IDs replace the numbering used in the older per-service reports in docs/.

1.2 Priority rubric

Priority	Meaning	Applied to
P1 - Must Pass	Release-blocking	Integration and security tests; core business logic (auth, bookings, payments, tournaments, loyalty, notifications); cases rated High in the earlier service reports
P2 - Should Pass	Important, not release-blocking	Controller-layer unit tests, game/station/review management, scheduler; cases rated Medium in the earlier reports
P3 - Nice to Pass	Supporting behaviour	Exception-handler formatting, utility classes, HTTP client wrappers, email templates; cases rated Low

2. Requirement Traceability Summary

Requirement IDs follow the REQ-<SVC>-<NN> convention already used in docs/. The reference column shows the FR (functional requirement), UC (use case) or architecture item each requirement comes from. "Feature-level" means the code cites no FR number, so the requirement was named from the feature under test;

Api-gateway

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-GW-01	Public authentication routing (register, login, password reset, account reactivation) to user-service without requiring JWT	FR-01, FR-03, FR-04, FR-05	0	4	1	5	5
REQ-GW-02	Public browsing routing (station availability, tournament catalog) without requiring JWT	FR-06, FR-14	0	2	2	4	4
REQ-GW-04	Injection and downstream forwarding of X-User-Id, X-User-Role, X-User-Email, X-User-FullName, and X-Gateway-Secret	Arch 2.2	0	3	4	7	7
REQ-GW-05	Protected route enforcement (booking management, payment processing, user profiles, loyalty points)	FR-07, FR-10, FR-16, FR-19	0	2	0	2	2
REQ-GW-06	Immediate 401 rejection for missing, non-Bearer, expired, wrong-secret, or malformed JWT tokens	Security OWASP A07	0	6	5	11	11
REQ-GW-07	Operational monitoring routing (/actuator/health) accessible without authentication	Ops & Monitoring	0	1	0	1	1
REQ-GW-08	Filter ordering verification ensuring JWT filter runs prior to Netty routing filters (order = -1)	Architecture	1	0	1	2	2

User-service

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-USR-01	Customer registration with unique email	FR-01	7	3	0	10	10
REQ-USR-02	Email and password login; role-based JWT issuance	FR-02	6	1	0	7	7
REQ-USR-03	Phone-number login and E.164 phone normalisation	FR-03	9	3	0	12	12
REQ-USR-04	Password reset via emailed verification code	FR-04	19	8	0	27	27
REQ-USR-05	Password reset via SMS verification code	FR-05	4	1	0	5	5
REQ-USR-06	Account lockout after repeated failed logins	FR-06	9	1	0	10	10
REQ-USR-07	Profile viewing and self-service account management	Feature-level	12	11	0	23	23
REQ-USR-08	Account deactivation and reactivation	Feature-level	34	9	0	43	43
REQ-USR-09	Admin user management (list, search, status control)	Feature-level	3	14	0	17	17

REQ-USR-10	Gateway secret validation and role-based access control	Feature-level	0	0	7	7	7
REQ-USR-11	Standardised error responses	Feature-level	12	0	0	12	12
REQ-USR-12	Notification dispatch to notification-service	FR-04, FR-05	6	0	0	6	6

Booking-service

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-BKG-01	Station availability and slot lookup	FR-07	3	0	0	3	3
REQ-BKG-02	Customer self-service booking with automatic payment	FR-08	21	2	0	23	18
REQ-BKG-03	Admin walk-in booking on behalf of a customer	FR-09	2	2	0	4	4
REQ-BKG-04	Own booking history and admin booking listing	FR-10	10	1	0	11	11
REQ-BKG-05	Booking cancellation, rescheduling and compensation	FR-13	29	2	0	31	29
REQ-BKG-06	Game catalogue and station-game configuration	FR-30	29	0	0	29	29
REQ-BKG-07	Game station management (admin)	Feature-level	26	0	0	26	26
REQ-BKG-08	Station reviews	Feature-level	3	0	0	3	0
REQ-BKG-09	Inter-service clients (user, payment, loyalty, notification)	FR-14	17	0	0	17	17
REQ-BKG-10	Gateway secret validation and role-based access control	Feature-level	0	0	7	7	7
REQ-BKG-11	Standardised error responses	Feature-level	9	0	0	9	9

Payment-service

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-PAY-01	Online & card payment processing with idempotency	-	26	3	0	29	17
REQ-PAY-02	Loyalty point reservation, confirmation, release, and award	-	13	1	0	14	14
REQ-PAY-03	Cash payment handling for bookings and tournaments	-	7	2	0	9	9
REQ-PAY-04	Customer payment receipts with ownership validation	-	6	3	0	9	9
REQ-PAY-05	Customer transaction history (/api/payments/mine)	-	2	1	0	3	3
REQ-PAY-06	Admin paged and filtered payment log (G7)	-	2	1	0	3	3
REQ-PAY-07	Admin dashboard payment statistics and trend (G11)	-	2	1	0	3	3
REQ-PAY-08	Admin CSV export of payments (G13)	-	1	1	0	2	2
REQ-PAY-09	Internal payment processing for microservices	-	2	2	0	4	4
REQ-PAY-10	API Gateway security, secret validation, and RBAC	-	2	6	10	18	18

Tournament-service

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-TRN-01	Admin creates and publishes tournament with validation	FR-13	3	2	0	5	5
REQ-TRN-02	Public guest & customer browsing of tournaments & details	FR-14	6	3	0	9	9
REQ-TRN-03	Admin updates tournament status (UPCOMING, ONGOING, COMPLETED)	FR-17	3	1	0	4	4
REQ-TRN-04	Admin cancels tournament with notification fanout	UC-09	5	1	0	6	6
REQ-TRN-05	Customer registration with capacity, deadline, payment charge & email confirmation	FR-14, UC-08, BR-06	24	4	0	28	28
REQ-TRN-06	Admin monitors registrations in real time	FR-15, UC-09	3	1	0	4	4
REQ-TRN-07	Admin generates single-elimination tournament bracket with byes	FR-15, FR-18	7	2	0	9	9
REQ-TRN-08	View full bracket with all rounds and matches	FR-15, FR-16	3	1	0	4	4
REQ-TRN-09	Admin records match result, auto-advances round, and crowns champion	FR-15, FR-18	10	1	0	11	11
REQ-TRN-10	Admin dashboard tournament stats widget	G11	2	1	0	3	3
REQ-TRN-11	Pre-tournament 2-hour reminder scheduler and manual reminder notification	-	11	1	0	12	12
REQ-TRN-12	API Gateway security, secret validation, and Role-Based Access Control (RBAC)	-	2	8	9	19	19

Loyalty-service

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-LOY-01	Points balance, tier status and transaction history	Feature-level	12	8	0	20	20
REQ-LOY-02	Points earning with tier multipliers; idempotent award	Feature-level	16	3	0	19	19
REQ-LOY-03	Points redemption: calculate, reserve, confirm, release	UC-11	19	7	0	26	26
REQ-LOY-04	Points reversal on refund or cancellation	Feature-level	5	0	0	5	5
REQ-LOY-05	Admin loyalty rules, adjustments, accounts and statistics	Feature-level	15	9	0	24	24
REQ-LOY-06	Expired reservation clean-up scheduler	Feature-level	4	0	0	4	4
REQ-LOY-07	Gateway secret validation and role-based access control	Feature-level	0	0	11	11	11
REQ-LOY-08	Standardised error responses	Feature-level	14	0	0	14	14

Notification-service (port 8086)

Requirement	Description	Reference	Unit	Int.	Sec.	Total	Pass
REQ-NTF-01	Account emails: welcome, password reset, reactivation	FR-04	7	3	0	10	10
REQ-NTF-02	Booking emails: confirmed, rescheduled, cancelled	Feature-level	6	3	0	9	9
REQ-NTF-03	Tournament emails: registration, cancellation, results, reminder	Feature-level	9	4	0	13	12
REQ-NTF-04	SMS delivery via Twilio (password reset, reactivation)	FR-05	10	2	0	12	12
REQ-NTF-05	Email template composition and SMTP transport	FR-04	12	0	0	12	12
REQ-NTF-06	Notification audit records (sent / failed)	Feature-level	3	0	0	3	3
REQ-NTF-07	Real-time socket broadcast	Feature-level	4	0	0	4	3
REQ-NTF-08	Gateway secret validation and service-role access control	Feature-level	0	2	5	7	7

3. Detailed Test Case Register

Tests are listed per service and per test class. Test File paths are relative to <service>/src/test/java/com/gamingcastle/<service-package>/. Status, Actual Output and any other cell can be edited directly after a new test run.

3.1 api-gateway - 32 test cases

ApiGatewayApplicationTest - 1 test(s)

Test Case ID	**TC-GW-UNIT-001**
Test Case Name	Application context loads successfully
Related Req.	REQ-GW-08
Related Test File	ApiGatewayApplicationTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Gateway Spring Boot reactive context boots successfully
Actual Output	Matches expected
Status	**Pass**

ApiGatewayIntegrationTest - 18 test(s)

Test Case ID	**TC-GW-INT-001**
Test Case Name	POST /api/auth/login routes to user-service without requiring auth header
Related Req.	REQ-GW-01
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/auth/login: 200 OK; Downstream received call without X-User-Id
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-002**
Test Case Name	POST /api/auth/register routes to user-service without requiring auth header
Related Req.	REQ-GW-01
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/auth/register: 201 CREATED; Downstream received register payload
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-003**
Test Case Name	POST /api/auth/password/forgot routes without requiring auth header
Related Req.	REQ-GW-01
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/auth/password/forgot: 200 OK; Downstream received forgot-password payload
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-004**
Test Case Name	POST /api/auth/reactivate/request routes without requiring auth header
Related Req.	REQ-GW-01
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/auth/reactivate/request: 200 OK; Downstream received reactivate payload
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-005**
Test Case Name	GET /api/stations routes to booking-service without requiring auth header
Related Req.	REQ-GW-02
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/stations: 200 OK; Downstream received call without X-User-Id
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-006**
Test Case Name	GET /api/tournaments routes to tournament-service without requiring auth header
Related Req.	REQ-GW-02
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/tournaments: 200 OK; Downstream received call without X-User-Id
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-007
Test Case Name	GET /api/tournaments with valid Bearer token forwards extracted headers downstream
Related Req.	REQ-GW-04
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/tournaments: 200 OK; Downstream received X-User-Id, X-User-Role, X-Gateway-Secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-008
Test Case Name	GET /api/tournaments with invalid Bearer token proceeds anonymously without error
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/tournaments: 200 OK; Downstream received call anonymously without X-User-Id
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-009
Test Case Name	GET /api/bookings/mine without token returns 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/bookings/mine: 401 UNAUTHORIZED; Blocked by gateway, downstream never called
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-010
Test Case Name	GET /api/bookings/mine with non-Bearer header returns 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/bookings/mine: 401 UNAUTHORIZED; Blocked by gateway, downstream never called
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-011**
Test Case Name	POST /api/payments with expired token returns 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/payments: 401 UNAUTHORIZED; Blocked by gateway, downstream never called
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-012**
Test Case Name	POST /api/payments with token signed with wrong secret returns 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/payments: 401 UNAUTHORIZED; Blocked by gateway, downstream never called
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-INT-013**
Test Case Name	POST /api/payments with malformed token returns 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/payments: 401 UNAUTHORIZED; Blocked by gateway, downstream never called
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-014
Test Case Name	GET /api/bookings/mine with valid CUSTOMER JWT routes to booking-service with mutated headers
Related Req.	REQ-GW-04
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/bookings/mine: 200 OK; Downstream received X-User-Id, X-User-Role: CUSTOMER, secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-015
Test Case Name	POST /api/bookings/admin/slots with valid ADMIN JWT routes with X-User-Role: ADMIN
Related Req.	REQ-GW-04
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/bookings/admin/slots: 200 OK; Downstream received X-User-Id, X-User-Role: ADMIN, secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-016
Test Case Name	GET /api/users/profile with valid token routes to user-service
Related Req.	REQ-GW-05
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/users/profile: 200 OK; Routed to user-service with headers
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-017
Test Case Name	GET /api/loyalty/points with valid token routes to loyalty-service
Related Req.	REQ-GW-05
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/loyalty/points: 200 OK; Routed to loyalty-service with headers
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-INT-018
Test Case Name	GET /actuator/health returns 200 OK without requiring authentication
Related Req.	REQ-GW-07
Related Test File	ApiGatewayIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /actuator/health: 200 OK; Handled by Actuator endpoint, returns status UP
Actual Output	Matches expected
Status	**Pass**

JwtAuthenticationFilterTest - 13 test(s)

Test Case ID	TC-GW-SEC-001
Test Case Name	getOrder should return -1 to execute before routing filters
Related Req.	REQ-GW-08
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Before routing filters
Expected Output	Filter returns -1
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-002
Test Case Name	Public paths without Authorization header should proceed anonymously
Related Req.	REQ-GW-01
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Request proceeds through filter chain without 401
Actual Output	Matches expected (10 variants)
Status	**Pass**

Test Case ID	TC-GW-SEC-003
Test Case Name	Public path with valid Bearer token should extract claims and forward headers
Related Req.	REQ-GW-04
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Claims extracted, mutated headers forwarded
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-004
Test Case Name	Public path with valid Bearer token having null email and fullName should forward empty strings
Related Req.	REQ-GW-04
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P2
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Null email & fullName fallback to empty string
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-005
Test Case Name	Public path with non-Bearer Authorization header should proceed anonymously without error
Related Req.	REQ-GW-02
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P2
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Non-Bearer header on public path proceeds anonymously
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-006
Test Case Name	Public path with invalid/expired token should continue anonymously without throwing or 401
Related Req.	REQ-GW-02
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P2
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Exception caught, request proceeds anonymously
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-007
Test Case Name	Protected path without Authorization header should return 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 "Missing or malformed Authorization header"
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-008
Test Case Name	Protected path with non-Bearer Authorization header should return 401 Unauthorized
Related Req.	REQ-GW-06
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 "Missing or malformed Authorization header"
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-009
Test Case Name	Protected path with expired JWT token should return 401 Invalid or expired token
Related Req.	REQ-GW-06
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 "Invalid or expired token"
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-010
Test Case Name	Protected path with JWT signed with wrong secret key should return 401 Invalid or expired token
Related Req.	REQ-GW-06
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 "Invalid or expired token"
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-GW-SEC-011
Test Case Name	Protected path with malformed JWT string should return 401 Invalid or expired token
Related Req.	REQ-GW-06
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 "Invalid or expired token"
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-SEC-012**
Test Case Name	Protected path with valid JWT should extract claims and forward headers
Related Req.	REQ-GW-04
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Headers forwarded with subject, role, email, and secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-GW-SEC-013**
Test Case Name	Protected path with valid JWT having null email and fullName should forward empty strings
Related Req.	REQ-GW-04
Related Test File	filter/ JwtAuthenticationFilterTest
Type	Security
Pri.	P2
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Fallback empty string headers forwarded
Actual Output	Matches expected
Status	**Pass**

3.2 user-service - 179 test cases

NotificationClientTest - 6 test(s)

Test Case ID	TC-USR-UNIT-001
Test Case Name	sendWelcomeEmail: should dispatch POST with gateway headers and body
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With gateway headers and body
Expected Output	Dispatch POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-002
Test Case Name	sendPasswordResetEmail: should dispatch POST with code payload
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With code payload
Expected Output	Dispatch POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-003
Test Case Name	sendPasswordResetSms: should dispatch POST with SMS payload
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With SMS payload
Expected Output	Dispatch POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-004
Test Case Name	sendAccountReactivationEmail: should dispatch POST with reactivation payload
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With reactivation payload
Expected Output	Dispatch POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-005
Test Case Name	sendAccountReactivationSms: should dispatch POST with reactivation SMS payload
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With reactivation SMS payload
Expected Output	Dispatch POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-006
Test Case Name	send: should catch exceptions gracefully when server returns 500 error
Related Req.	REQ-USR-12
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	When server returns 500 error
Expected Output	Catch exceptions gracefully
Actual Output	Matches expected
Status	**Pass**

AuthControllerTest - 19 test(s)

Test Case ID	TC-USR-UNIT-007
Test Case Name	POST /api/auth/register: should return 201 Created and AuthResponse given valid request
Related Req.	REQ-USR-01
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid request
Expected Output	Return 201 Created and AuthResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-008
Test Case Name	POST /api/auth/register: should return 400 Bad Request given invalid email
Related Req.	REQ-USR-01
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given invalid email
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-009
Test Case Name	POST /api/auth/register: should return 400 Bad Request given password shorter than 8 chars
Related Req.	REQ-USR-01
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given password shorter than 8 chars
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-010
Test Case Name	POST /api/auth/register: should return 409 Conflict when email is already taken
Related Req.	REQ-USR-01
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When email is already taken
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-011
Test Case Name	POST /api/auth/login: should return 200 OK and AuthResponse given valid credentials
Related Req.	REQ-USR-02
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid credentials
Expected Output	Return 200 OK and AuthResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-012
Test Case Name	POST /api/auth/login: should return 400 Bad Request given blank password
Related Req.	REQ-USR-02
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given blank password
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-013**
Test Case Name	POST /api/auth/login: should return 401 Unauthorized given invalid credentials
Related Req.	REQ-USR-02
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given invalid credentials
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-014**
Test Case Name	POST /api/auth/login: should return 423 Locked with Retry-After header when account is locked
Related Req.	REQ-USR-06
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	With Retry-After header when account is locked
Expected Output	Return 423 Locked
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-015**
Test Case Name	POST /api/auth/login/phone: should return 200 OK and AuthResponse given valid phone credentials
Related Req.	REQ-USR-03
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid phone credentials
Expected Output	Return 200 OK and AuthResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-016**
Test Case Name	POST /api/auth/login/phone: should return 400 Bad Request given blank phone number
Related Req.	REQ-USR-03
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given blank phone number
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-017
Test Case Name	POST /api/auth/password/forgot: should return 202 Accepted given valid identifier
Related Req.	REQ-USR-04
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid identifier
Expected Output	Return 202 Accepted
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-018
Test Case Name	POST /api/auth/password/forgot: should return 400 Bad Request given blank identifier
Related Req.	REQ-USR-04
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given blank identifier
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-019
Test Case Name	POST /api/auth/password/reset: should return 200 OK given valid reset request
Related Req.	REQ-USR-04
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid reset request
Expected Output	Return 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-020
Test Case Name	POST /api/auth/password/reset: should return 400 Bad Request when reset code is invalid
Related Req.	REQ-USR-04
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When reset code is invalid
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-021
Test Case Name	POST /api/auth/password/reset: should return 400 Bad Request when new password is too short
Related Req.	REQ-USR-04
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When new password is too short
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-022
Test Case Name	POST /api/auth/reactivate/request: should return 202 Accepted given valid identifier
Related Req.	REQ-USR-08
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid identifier
Expected Output	Return 202 Accepted
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-023
Test Case Name	POST /api/auth/reactivate/request: should return 400 Bad Request given blank identifier
Related Req.	REQ-USR-08
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given blank identifier
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-024
Test Case Name	POST /api/auth/reactivate/confirm: should return 200 OK and AuthResponse given valid code
Related Req.	REQ-USR-08
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid code
Expected Output	Return 200 OK and AuthResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-025
Test Case Name	POST /api/auth/reactivate/confirm: should return 400 Bad Request when code is invalid
Related Req.	REQ-USR-08
Related Test File	controller/ AuthControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When code is invalid
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

UserControllerTest - 14 test(s)

Test Case ID	TC-USR-UNIT-026
Test Case Name	GET /api/users/me: should return 200 OK and UserProfileResponse given valid X-User-Id header
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid X-User-Id header
Expected Output	Return 200 OK and UserProfileResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-027
Test Case Name	GET /api/users/me: should return 400 Bad Request when X-User-Id header is missing
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When X-User-Id header is missing
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-028
Test Case Name	GET /api/users/me: should return 404 Not Found when user does not exist
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When user does not exist
Expected Output	Return 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-029**
Test Case Name	PATCH /api/users/me: should return 200 OK and updated UserProfileResponse given valid request
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given valid request
Expected Output	Return 200 OK and updated UserProfileResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-030**
Test Case Name	PATCH /api/users/me: should return 400 Bad Request when fullName is blank
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When fullName is blank
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-031**
Test Case Name	GET /api/users: should return 200 OK and paged user summaries
Related Req.	REQ-USR-09
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Return 200 OK and paged user summaries
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-032**
Test Case Name	GET /api/users/{userId}: should return 200 OK and UserSummaryResponse given existing user
Related Req.	REQ-USR-09
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Given existing user
Expected Output	Return 200 OK and UserSummaryResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-033
Test Case Name	GET /api/users/{userId}: should return 404 Not Found when user does not exist
Related Req.	REQ-USR-07
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When user does not exist
Expected Output	Return 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-034
Test Case Name	PATCH /api/users/me/deactivate: should return 204 No Content when user self-deactivates
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When user self-deactivates
Expected Output	Return 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-035
Test Case Name	PATCH /api/users/me/deactivate: should return 409 Conflict when account is already deactivated
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When account is already deactivated
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-036
Test Case Name	PATCH /api/users/{userId}/deactivate: should return 204 No Content when admin deactivates user
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When admin deactivates user
Expected Output	Return 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-037
Test Case Name	PATCH /api/users/{userId}/deactivate: should return 409 Conflict when target user is already deactivated
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When target user is already deactivated
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-038
Test Case Name	PATCH /api/users/{userId}/activate: should return 204 No Content when admin activates user
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When admin activates user
Expected Output	Return 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-039
Test Case Name	PATCH /api/users/{userId}/activate: should return 409 Conflict when user is already active
Related Req.	REQ-USR-08
Related Test File	controller/ UserControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When user is already active
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

GlobalExceptionHandlerTest - 12 test(s)

Test Case ID	TC-USR-UNIT-040
Test Case Name	handleValidation: should return 400 Bad Request with VALIDATION_ERROR and field errors map
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With VALIDATION_ERROR and field errors map
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-041
Test Case Name	handleUserNotFound: should return 404 Not Found with USER_NOT_FOUND
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With USER_NOT_FOUND
Expected Output	Return 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-042
Test Case Name	handleEmailAlreadyExists: should return 409 Conflict with EMAIL_TAKEN
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With EMAIL_TAKEN
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-043
Test Case Name	handleInvalidCredentials: should return 401 Unauthorized with INVALID_CREDENTIALS
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With INVALID_CREDENTIALS
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-044
Test Case Name	handleAccountLocked: should return 423 Locked with ACCOUNT_LOCKED and Retry-After header
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With ACCOUNT_LOCKED and Retry-After header
Expected Output	Return 423 Locked
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-045
Test Case Name	handlePhoneAlreadyExists: should return 409 Conflict with PHONE_TAKEN
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With PHONE_TAKEN
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-046
Test Case Name	handleInvalidResetCode: should return 400 Bad Request with INVALID_RESET_CODE
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With INVALID_RESET_CODE
Expected Output	Return 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-047
Test Case Name	handleNotificationDeliveryFailed: should return 502 Bad Gateway with NOTIFICATION_DELIVERY_FAILED
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With NOTIFICATION_DELIVERY_FAILED
Expected Output	Return 502 Bad Gateway
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-048
Test Case Name	handleUserAlreadyDeactivated: should return 409 Conflict with USER_ALREADY_DEACTIVATED
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With USER_ALREADY_DEACTIVATED
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-049
Test Case Name	handleAccountDeactivated: should return 403 Forbidden with ACCOUNT_DEACTIVATED
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With ACCOUNT_DEACTIVATED
Expected Output	Return 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-050
Test Case Name	handleAccountDeactivatedByAdmin: should return 403 Forbidden with ACCOUNT_DEACTIVATED_BY_ADMIN
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With ACCOUNT_DEACTIVATED_BY_ADMIN
Expected Output	Return 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-051
Test Case Name	handleUserAlreadyActive: should return 409 Conflict with USER_ALREADY_ACTIVE
Related Req.	REQ-USR-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With USER_ALREADY_ACTIVE
Expected Output	Return 409 Conflict
Actual Output	Matches expected
Status	**Pass**

AccountReactivationServiceTest - 8 test(s)

Test Case ID	TC-USR-UNIT-052
Test Case Name	Request reactivation should issue code given self deactivated user
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given self deactivated user
Expected Output	Issue code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-053
Test Case Name	Request reactivation should do nothing given admin deactivated user
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given admin deactivated user
Expected Output	Do nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-054
Test Case Name	Request reactivation should do nothing given active user
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given active user
Expected Output	Do nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-055
Test Case Name	Request reactivation should do nothing given unknown identifier
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown identifier
Expected Output	Do nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-056
Test Case Name	Confirm reactivation should activate and login given valid code
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given valid code
Expected Output	Activate and login
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-057
Test Case Name	Confirm reactivation should stay deactivated given invalid code
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given invalid code
Expected Output	Stay deactivated
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-058
Test Case Name	Confirm reactivation should reject given admin deactivated user
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given admin deactivated user
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-059
Test Case Name	Confirm reactivation should reject given unknown identifier
Related Req.	REQ-USR-08
Related Test File	service/ AccountReactivationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown identifier
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

AuthServiceTest - 19 test(s)

Test Case ID	TC-USR-UNIT-060
Test Case Name	Register should create customer account given new email
Related Req.	REQ-USR-01
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given new email
Expected Output	Create customer account
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-061
Test Case Name	Register should throw conflict given email already exists
Related Req.	REQ-USR-01
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given email already exists
Expected Output	Throw conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-062
Test Case Name	Login should return token given valid credentials
Related Req.	REQ-USR-02
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given valid credentials
Expected Output	Return token
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-063
Test Case Name	Login should lock account after five failed attempts
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	After five failed attempts
Expected Output	Lock account
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-064
Test Case Name	Login should reject locked account even with correct password
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Even with correct password
Expected Output	Reject locked account
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-065
Test Case Name	Login by phone should return token given valid credentials
Related Req.	REQ-USR-03
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given valid credentials
Expected Output	Return token
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-066
Test Case Name	Login by phone should reject invalid credentials given unknown phone number
Related Req.	REQ-USR-03
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown phone number
Expected Output	Reject invalid credentials
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-067
Test Case Name	Login should require verification given self deactivated account
Related Req.	REQ-USR-08
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given self deactivated account
Expected Output	Require verification
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-068
Test Case Name	Login should show admin message given admin deactivated account
Related Req.	REQ-USR-08
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given admin deactivated account
Expected Output	Show admin message
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-069
Test Case Name	Login should not reveal deactivation given wrong password
Related Req.	REQ-USR-08
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given wrong password
Expected Output	Not reveal deactivation
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-070
Test Case Name	Login by phone should require verification given self deactivated account
Related Req.	REQ-USR-08
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given self deactivated account
Expected Output	Require verification
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-071
Test Case Name	Login by phone should show admin message given admin deactivated account
Related Req.	REQ-USR-08
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given admin deactivated account
Expected Output	Show admin message
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-072
Test Case Name	Login should start counting from one when previous lock has expired
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When previous lock has expired
Expected Output	Start counting from one
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-073
Test Case Name	Login should expose lock end time when account is locked
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When account is locked
Expected Output	Expose lock end time
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-074
Test Case Name	Login by phone should lock account after five failed attempts
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	After five failed attempts
Expected Output	Lock account
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-075
Test Case Name	Register should throw conflict given phone already exists
Related Req.	REQ-USR-03
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given phone already exists
Expected Output	Throw conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-076
Test Case Name	Register should succeed with notification sent false when notification service fails
Related Req.	REQ-USR-01
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When notification service fails
Expected Output	Succeed with notification sent false
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-077
Test Case Name	Login should reject invalid credentials given unknown email
Related Req.	REQ-USR-02
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown email
Expected Output	Reject invalid credentials
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-078
Test Case Name	Login by phone should start counting from one when previous lock has expired
Related Req.	REQ-USR-06
Related Test File	service/ AuthServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When previous lock has expired
Expected Output	Start counting from one
Actual Output	Matches expected
Status	**Pass**

PasswordResetServiceTest - 6 test(s)

Test Case ID	TC-USR-UNIT-079
Test Case Name	Request reset should issue reset code given email identifier
Related Req.	REQ-USR-04
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given email identifier
Expected Output	Issue reset code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-080
Test Case Name	Request reset should issue reset code given phone identifier
Related Req.	REQ-USR-05
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given phone identifier
Expected Output	Issue reset code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-081
Test Case Name	Request reset should do nothing given unknown identifier
Related Req.	REQ-USR-04
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown identifier
Expected Output	Do nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-082
Test Case Name	Reset password should update password given valid code
Related Req.	REQ-USR-04
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given valid code
Expected Output	Update password
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-083
Test Case Name	Reset password should reject given invalid code
Related Req.	REQ-USR-04
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given invalid code
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-084
Test Case Name	Reset password should reject given unknown identifier
Related Req.	REQ-USR-04
Related Test File	service/ PasswordResetServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown identifier
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

TransactionRollbackRulesTest - 2 test(s)

Test Case ID	**TC-USR-UNIT-085**
Test Case Name	Login should commit failed attempt when password is wrong
Related Req.	REQ-USR-06
Related Test File	service/ TransactionRollbackRulesTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When password is wrong
Expected Output	Commit failed attempt
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-086**
Test Case Name	Login should commit lock when fifth wrong password triggers lockout
Related Req.	REQ-USR-06
Related Test File	service/ TransactionRollbackRulesTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When fifth wrong password triggers lockout
Expected Output	Commit lock
Actual Output	Matches expected
Status	**Pass**

UserServiceTest - 18 test(s)

Test Case ID	**TC-USR-UNIT-087**
Test Case Name	Deactivate my account should mark as self
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Mark as self
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-088**
Test Case Name	Deactivate my account should reject when already deactivated
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When already deactivated
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-089
Test Case Name	Deactivate user should mark as admin when admin deactivates someone else
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When admin deactivates someone else
Expected Output	Mark as admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-090
Test Case Name	Deactivate user should mark as self when admin deactivates their own account
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When admin deactivates their own account
Expected Output	Mark as self
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-091
Test Case Name	Deactivate user should upgrade self deactivation to admin
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Upgrade self deactivation to admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-092
Test Case Name	Deactivate user should reject when already deactivated by admin
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When already deactivated by admin
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-093
Test Case Name	Deactivate user should throw not found given unknown user
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown user
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-094
Test Case Name	Activate user should reactivate and clear deactivation info
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Reactivate and clear deactivation info
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-095
Test Case Name	Activate user should reject when already active
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When already active
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-096
Test Case Name	Get my profile should return profile when user exists
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user exists
Expected Output	Return profile
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-097
Test Case Name	Get my profile should throw not found when user does not exist
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user does not exist
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-098
Test Case Name	Update my profile should update only full name and phone number when user exists
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user exists
Expected Output	Update only full name and phone number
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-099
Test Case Name	Update my profile should throw not found when user does not exist
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user does not exist
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-100
Test Case Name	Get all users should return paged user summaries given pageable
Related Req.	REQ-USR-09
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given pageable
Expected Output	Return paged user summaries
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-101
Test Case Name	Get user by id should return summary when user exists
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user exists
Expected Output	Return summary
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-102
Test Case Name	Get user by id should throw not found when user does not exist
Related Req.	REQ-USR-07
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user does not exist
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-103
Test Case Name	Deactivate my account should throw not found given unknown user
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown user
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-104
Test Case Name	Activate user should throw not found given unknown user
Related Req.	REQ-USR-08
Related Test File	service/ UserServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given unknown user
Expected Output	Throw not found
Actual Output	Matches expected
Status	**Pass**

VerificationCodeServiceTest - 12 test(s)

Test Case ID	**TC-USR-UNIT-105**
Test Case Name	Find user by identifier should look up by email when identifier contains at
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When identifier contains at
Expected Output	Look up by email
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-106**
Test Case Name	Find user by identifier should look up by phone when no at
Related Req.	REQ-USR-05
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When no at
Expected Output	Look up by phone
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-107**
Test Case Name	Issue code should email account reactivation code given email identifier and reactivation purpose
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given email identifier and reactivation purpose
Expected Output	Email account reactivation code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-108**
Test Case Name	Issue code should email password reset code given email identifier and password reset purpose
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given email identifier and password reset purpose
Expected Output	Email password reset code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-109
Test Case Name	Issue code should text password reset code given phone identifier and password reset purpose
Related Req.	REQ-USR-05
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given phone identifier and password reset purpose
Expected Output	Text password reset code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-110
Test Case Name	Issue code should text account reactivation code given phone identifier and reactivation purpose
Related Req.	REQ-USR-05
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given phone identifier and reactivation purpose
Expected Output	Text account reactivation code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-111
Test Case Name	Verify and consume should mark code used given valid code
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given valid code
Expected Output	Mark code used
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-112
Test Case Name	Verify and consume should reject given expired code
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given expired code
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-113
Test Case Name	Verify and consume should reject given wrong code
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Given wrong code
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-114
Test Case Name	Verify and consume should reject when no code exists for that purpose
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When no code exists for that purpose
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-115
Test Case Name	Verify and consume should burn code after five wrong guesses
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	After five wrong guesses
Expected Output	Burn code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-116
Test Case Name	Issue code should not send new code when resend limit reached
Related Req.	REQ-USR-04
Related Test File	service/ VerificationCodeServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When resend limit reached
Expected Output	Not send new code
Actual Output	Matches expected
Status	**Pass**

JwtUtilTest - 1 test(s)

Test Case ID	**TC-USR-UNIT-117**
Test Case Name	generateToken: should generate a valid HMAC-SHA signed JWT with subject, email, fullName, and role
Related Req.	REQ-USR-02
Related Test File	util/ JwtUtilTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With subject, email, fullName, and role
Expected Output	Generate a valid HMAC-SHA signed JWT
Actual Output	Matches expected
Status	**Pass**

PhoneNumberUtilTest - 4 test(s)

Test Case ID	**TC-USR-UNIT-118**
Test Case Name	toE164: should keep number unchanged when already in international format starting with '+'
Related Req.	REQ-USR-03
Related Test File	util/ PhoneNumberUtilTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	When already in international format starting with '+'
Expected Output	Keep number unchanged
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-119**
Test Case Name	toE164: should replace leading zero with default country code
Related Req.	REQ-USR-03
Related Test File	util/ PhoneNumberUtilTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	With default country code
Expected Output	Replace leading zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-UNIT-120**
Test Case Name	toE164: should prepend country code when number starts without zero or plus
Related Req.	REQ-USR-03
Related Test File	util/ PhoneNumberUtilTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	When number starts without zero or plus
Expected Output	Prepend country code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-UNIT-121
Test Case Name	toE164: should remove spaces, hyphens and whitespace
Related Req.	REQ-USR-03
Related Test File	util/ PhoneNumberUtilTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Remove spaces, hyphens and whitespace
Actual Output	Matches expected (3 variants)
Status	**Pass**

UserServiceAdminIntegrationTest - 14 test(s)

Test Case ID	TC-USR-INT-001
Test Case Name	GET /api/users - admin caller returns 200 OK with paginated user summaries
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users; admin caller
Expected Output	Returns 200 OK with paginated user summaries
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-002
Test Case Name	GET /api/users - customer caller is denied with 403 Forbidden
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users; customer caller
Expected Output	Is denied with 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-003
Test Case Name	GET /api/users/{userId} - admin caller returns 200 OK with user summary
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/{userId}; admin caller
Expected Output	Returns 200 OK with user summary
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-004**
Test Case Name	GET /api/users/{userId} - admin caller querying non-existent user returns 404 Not Found
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/{userId}; admin caller querying non-existent user
Expected Output	Returns 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-005**
Test Case Name	GET /api/users/{userId} - customer caller is denied with 403 Forbidden
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/{userId}; customer caller
Expected Output	Is denied with 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-006**
Test Case Name	PATCH /api/users/{userId}/deactivate - admin deactivates active target returning 204 No Content
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin deactivates active target returning 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-007**
Test Case Name	PATCH /api/users/{userId}/deactivate - admin takes over self-deactivated account returning 204
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin takes over self-deactivated account returning 204
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-008
Test Case Name	PATCH /api/users/{userId}/deactivate - deactivating already admin-deactivated user returns 409 Conflict
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/deactivate; deactivating already admin-deactivated user
Expected Output	Returns 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-009
Test Case Name	PATCH /api/users/{userId}/deactivate - deactivating unknown user returns 404 Not Found
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/deactivate; deactivating unknown user
Expected Output	Returns 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-010
Test Case Name	PATCH /api/users/{userId}/deactivate - customer caller is denied with 403 Forbidden
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/deactivate; customer caller
Expected Output	Is denied with 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-011
Test Case Name	PATCH /api/users/{userId}/activate - admin reactivates deactivated target returning 204 No Content
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin reactivates deactivated target returning 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-012
Test Case Name	PATCH /api/users/{userId}/activate - activating already active user returns 409 Conflict
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/activate; activating already active user
Expected Output	Returns 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-013
Test Case Name	PATCH /api/users/{userId}/activate - activating unknown user returns 404 Not Found
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/activate; activating unknown user
Expected Output	Returns 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-014
Test Case Name	PATCH /api/users/{userId}/activate - customer caller is denied with 403 Forbidden
Related Req.	REQ-USR-09
Related Test File	controller/ UserServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/{userId}/activate; customer caller
Expected Output	Is denied with 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

UserServiceAuthIntegrationTest - 26 test(s)

Test Case ID	TC-USR-INT-015
Test Case Name	POST /api/auth/register - successful registration persists user and returns 201 with JWT
Related Req.	REQ-USR-01
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/register; successful registration persists user and
Expected Output	Returns 201 with JWT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-016**
Test Case Name	POST /api/auth/register - duplicate email returns 409 Conflict with EMAIL_TAKEN
Related Req.	REQ-USR-01
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/register; duplicate email
Expected Output	Returns 409 Conflict with EMAIL_TAKEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-017**
Test Case Name	POST /api/auth/register - duplicate phone returns 409 Conflict with PHONE_TAKEN
Related Req.	REQ-USR-03
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/register; duplicate phone
Expected Output	Returns 409 Conflict with PHONE_TAKEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-018**
Test Case Name	POST /api/auth/register - validation failure returns 400 Bad Request with field errors
Related Req.	REQ-USR-01
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/register; validation failure
Expected Output	Returns 400 Bad Request with field errors
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-019**
Test Case Name	POST /api/auth/login - successful login returns 200 OK with valid JWT and user info
Related Req.	REQ-USR-02
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login; successful login
Expected Output	Returns 200 OK with valid JWT and user info
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-020**
Test Case Name	POST /api/auth/login - self-deactivated account returns 403 Forbidden with ACCOUNT_DEACTIVATED
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login; self-deactivated account
Expected Output	Returns 403 Forbidden with ACCOUNT_DEACTIVATED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-021**
Test Case Name	POST /api/auth/login - admin-deactivated account returns 403 Forbidden with ACCOUNT_DEACTIVATED_BY_ADMIN
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login; admin-deactivated account
Expected Output	Returns 403 Forbidden with ACCOUNT_DEACTIVATED_BY_ADMIN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-022**
Test Case Name	POST /api/auth/login - lockout after 5 failed attempts returns 423 Locked with Retry-After header
Related Req.	REQ-USR-06
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login; lockout after 5 failed attempts
Expected Output	Returns 423 Locked with Retry-After header
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-023**
Test Case Name	POST /api/auth/login/phone - successful phone login returns 200 OK with valid JWT
Related Req.	REQ-USR-03
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login/phone; successful phone login
Expected Output	Returns 200 OK with valid JWT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-024
Test Case Name	POST /api/auth/login/phone - invalid phone or wrong password returns 401 Unauthorized
Related Req.	REQ-USR-03
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/login/phone; invalid phone or wrong password
Expected Output	Returns 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-025
Test Case Name	POST /api/auth/password/forgot - known email returns 202 Accepted and issues email verification code
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/forgot; known email
Expected Output	Returns 202 Accepted and issues email verification code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-026
Test Case Name	POST /api/auth/password/forgot - known phone returns 202 Accepted and issues SMS verification code
Related Req.	REQ-USR-05
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/forgot; known phone
Expected Output	Returns 202 Accepted and issues SMS verification code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-027
Test Case Name	POST /api/auth/password/forgot - blank identifier returns 400 Bad Request with VALIDATION_ERROR
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/forgot; blank identifier
Expected Output	Returns 400 Bad Request with VALIDATION_ERROR
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-028**
Test Case Name	POST /api/auth/password/forgot - rate limit exceeded prevents issuing more than 3 codes per window
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Rate limit exceeded prevents issuing more than 3 codes per window
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-029**
Test Case Name	POST /api/auth/password/reset - valid OTP updates password, resets lockout, and returns 200 OK
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/reset; valid OTP updates password, resets lockout, and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-030**
Test Case Name	POST /api/auth/password/reset - invalid OTP returns 400 Bad Request and increments failedAttempts
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/reset; invalid OTP
Expected Output	Returns 400 Bad Request and increments failedAttempts
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-031**
Test Case Name	POST /api/auth/password/reset - expired OTP returns 400 Bad Request with INVALID_RESET_CODE
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/reset; expired OTP
Expected Output	Returns 400 Bad Request with INVALID_RESET_CODE
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-032
Test Case Name	POST /api/auth/password/reset - burned OTP after 5 failed attempts rejects further reset attempts
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Burned OTP after 5 failed attempts rejects further reset attempts
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-033
Test Case Name	POST /api/auth/password/reset - validation failure returns 400 Bad Request with field errors
Related Req.	REQ-USR-04
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/password/reset; validation failure
Expected Output	Returns 400 Bad Request with field errors
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-034
Test Case Name	POST /api/auth/reactivate/request - self-deactivated account returns 202 Accepted and issues code
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/request; self-deactivated account
Expected Output	Returns 202 Accepted and issues code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-035
Test Case Name	POST /api/auth/reactivate/request - unknown identifier returns 202 Accepted without issuing code
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/request; unknown identifier
Expected Output	Returns 202 Accepted without issuing code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-036**
Test Case Name	POST /api/auth/reactivate/request - blank identifier returns 400 Bad Request with VALIDATION_ERROR
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/request; blank identifier
Expected Output	Returns 400 Bad Request with VALIDATION_ERROR
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-037**
Test Case Name	POST /api/auth/reactivate/confirm - valid OTP reactivates account and returns 200 OK with JWT
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/confirm; valid OTP reactivates account and
Expected Output	Returns 200 OK with JWT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-038**
Test Case Name	POST /api/auth/reactivate/confirm - invalid OTP returns 400 Bad Request and account remains deactivated
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/confirm; invalid OTP
Expected Output	Returns 400 Bad Request and account remains deactivated
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-039**
Test Case Name	POST /api/auth/reactivate/confirm - admin-deactivated account returns 400 Bad Request with INVALID_RESET_CODE
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/confirm; admin-deactivated account
Expected Output	Returns 400 Bad Request with INVALID_RESET_CODE
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-040
Test Case Name	POST /api/auth/reactivate/confirm - validation failure returns 400 Bad Request with field errors
Related Req.	REQ-USR-08
Related Test File	controller/ UserServiceAuthIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/auth/reactivate/confirm; validation failure
Expected Output	Returns 400 Bad Request with field errors
Actual Output	Matches expected
Status	**Pass**

UserServiceProfileIntegrationTest - 11 test(s)

Test Case ID	TC-USR-INT-041
Test Case Name	GET /api/users/me - valid authenticated user returns 200 with full profile data
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/me; valid authenticated user
Expected Output	Returns 200 with full profile data
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-042
Test Case Name	GET /api/users/me - missing or invalid X-Gateway-Secret is rejected with 401 Unauthorized
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/me; missing or invalid X-Gateway-Secret
Expected Output	Is rejected with 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-043
Test Case Name	GET /api/users/me - missing or invalid X-User-Id is rejected with 401 Unauthorized
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/me; missing or invalid X-User-Id
Expected Output	Is rejected with 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-044**
Test Case Name	GET /api/users/me - unknown user id is rejected with 401 Unauthorized
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/me; unknown user id
Expected Output	Is rejected with 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-045**
Test Case Name	GET /api/users/me - deactivated user is rejected with 401 Unauthorized
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/users/me; deactivated user
Expected Output	Is rejected with 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-046**
Test Case Name	PATCH /api/users/me - valid fullName update returns 200, updates DB, and preserves other fields
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/me; valid fullName update
Expected Output	Returns 200, updates DB, and preserves other fields
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-INT-047**
Test Case Name	PATCH /api/users/me - optional phoneNumber update updates phone and preserves other fields
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Optional phoneNumber update updates phone and preserves other fields
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-048
Test Case Name	PATCH /api/users/me - blank fullName validation returns 400 Bad Request with field errors
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/me; blank fullName validation
Expected Output	Returns 400 Bad Request with field errors
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-049
Test Case Name	PATCH /api/users/me - null fullName validation returns 400 Bad Request with VALIDATION_ERROR
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/me; null fullName validation
Expected Output	Returns 400 Bad Request with VALIDATION_ERROR
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-050
Test Case Name	PATCH /api/users/me/deactivate - active user successfully self-deactivates returning 204 No Content
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Active user successfully self-deactivates returning 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-INT-051
Test Case Name	PATCH /api/users/me/deactivate - second self-deactivation is rejected and deactivated user cannot access protected endpoints
Related Req.	REQ-USR-07
Related Test File	controller/ UserServiceProfileIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PATCH /api/users/me/deactivate; second self-deactivation
Expected Output	Is rejected and deactivated user cannot access protected endpoints
Actual Output	Matches expected
Status	**Pass**

GatewayAuthenticationFilterTest - 7 test(s)

Test Case ID	**TC-USR-SEC-001**
Test Case Name	doFilter: should bypass public paths without gateway secret or user headers
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Bypass public paths without gateway secret or user headers
Actual Output	Matches expected (10 variants)
Status	**Pass**

Test Case ID	**TC-USR-SEC-002**
Test Case Name	doFilter: should return 401 Unauthorized when X-Gateway-Secret header is missing
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When X-Gateway-Secret header is missing
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-SEC-003**
Test Case Name	doFilter: should return 401 Unauthorized when X-Gateway-Secret header is invalid
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When X-Gateway-Secret header is invalid
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-USR-SEC-004**
Test Case Name	doFilter: should return 401 Unauthorized when X-User-Id header is missing
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When X-User-Id header is missing
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-SEC-005
Test Case Name	doFilter: should return 401 Unauthorized when X-User-Id header is not a valid UUID
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When X-User-Id header is not a valid UUID
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-SEC-006
Test Case Name	doFilter: should return 401 Unauthorized when user is not found in database
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When user is not found in database
Expected Output	Return 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-USR-SEC-007
Test Case Name	doFilter: should authenticate user, set SecurityContext with DB role, and clear context in finally
Related Req.	REQ-USR-10
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	With DB role, and clear context in finally
Expected Output	Authenticate user, set SecurityContext
Actual Output	Matches expected
Status	**Pass**

3.3 booking-service - 163 test cases

LoyaltyClientTest - 3 test(s)

Test Case ID	TC-BKG-UNIT-001
Test Case Name	reverse: succeeds on first attempt with 200 OK and headers
Related Req.	REQ-BKG-09
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Succeeds on first attempt with 200 OK and headers
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-002
Test Case Name	reverse: retries once on server error and succeeds on second attempt
Related Req.	REQ-BKG-09
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Retries once on server error and succeeds on second attempt
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-003
Test Case Name	reverse: logs error and completes cleanly when both attempts fail
Related Req.	REQ-BKG-09
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Logs error and completes cleanly when both attempts fail
Actual Output	Matches expected
Status	**Pass**

NotificationClientTest - 6 test(s)

Test Case ID	TC-BKG-UNIT-004
Test Case Name	sendBookingConfirmation: dispatches POST with gateway headers and body
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST with gateway headers and body
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-005
Test Case Name	sendBookingRescheduled: dispatches POST with gateway headers and body
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST with gateway headers and body
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-006
Test Case Name	sendBookingCancelled: dispatches POST with gateway headers and body
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST with gateway headers and body
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-007
Test Case Name	sendBookingConfirmation: swallows network error without throwing exception
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Swallows network error without throwing exception
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-008
Test Case Name	sendBookingRescheduled: swallows network error without throwing exception
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Swallows network error without throwing exception
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-009
Test Case Name	sendBookingCancelled: swallows network error without throwing exception
Related Req.	REQ-BKG-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Swallows network error without throwing exception
Actual Output	Matches expected
Status	**Pass**

PaymentClientTest - 5 test(s)

Test Case ID	TC-BKG-UNIT-010
Test Case Name	chargeBookingFee: successful charge with explicit method and loyalty points
Related Req.	REQ-BKG-09
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successful charge with explicit method and loyalty points
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-011
Test Case Name	chargeBookingFee: defaults to CARD method when method is null or blank, handles null secret
Related Req.	REQ-BKG-09
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Defaults to CARD method when method is null or blank, handles null secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-012
Test Case Name	chargeBookingFee: throws 402 PAYMENT_REQUIRED when response body is null
Related Req.	REQ-BKG-09
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 402 PAYMENT_REQUIRED when response body is null
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-013
Test Case Name	chargeBookingFee: throws 402 PAYMENT_REQUIRED when payment status is not SUCCESS
Related Req.	REQ-BKG-09
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 402 PAYMENT_REQUIRED when payment status is not SUCCESS
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-014
Test Case Name	chargeBookingFee: throws 402 PAYMENT_REQUIRED on 5xx or connection error
Related Req.	REQ-BKG-09
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 402 PAYMENT_REQUIRED on 5xx or connection error
Actual Output	Matches expected
Status	**Pass**

UserClientTest - 3 test(s)

Test Case ID	TC-BKG-UNIT-015
Test Case Name	getUserById: returns user summary when user service responds 200 OK
Related Req.	REQ-BKG-09
Related Test File	client/ UserClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns user summary when user service responds 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-016
Test Case Name	getUserById: omits X-Gateway-Secret when blank
Related Req.	REQ-BKG-09
Related Test File	client/ UserClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Omits X-Gateway-Secret when blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-017
Test Case Name	getUserById: returns empty Optional when service returns error
Related Req.	REQ-BKG-09
Related Test File	client/ UserClientTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	getUserById; returns empty Optional when service
Expected Output	Returns error
Actual Output	Matches expected
Status	**Pass**

BookingControllerTest - 19 test(s)

Test Case ID	TC-BKG-UNIT-018
Test Case Name	POST /api/bookings: successfully creates online booking
Related Req.	REQ-BKG-02
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Successfully creates online booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-019
Test Case Name	POST /api/bookings: returns 400 when validation fails on null stationId
Related Req.	REQ-BKG-02
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when validation fails on null stationId
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-020
Test Case Name	POST /api/bookings: returns 409 when slot is unavailable
Related Req.	REQ-BKG-01
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 409 when slot is unavailable
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-021
Test Case Name	POST /api/bookings: returns 402 when payment fails
Related Req.	REQ-BKG-02
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 402 when payment fails
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-022
Test Case Name	POST /api/bookings/walk-in: successfully creates walk-in booking as admin
Related Req.	REQ-BKG-03
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Successfully creates walk-in booking as admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-023
Test Case Name	POST /api/bookings/walk-in: returns 400 when targetUserId is null
Related Req.	REQ-BKG-03
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when targetUserId is null
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-024
Test Case Name	GET /api/bookings/mine: returns list of user bookings
Related Req.	REQ-BKG-04
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns list of user bookings
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-025
Test Case Name	GET /api/bookings/mine: returns empty list when no bookings exist
Related Req.	REQ-BKG-04
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns empty list when no bookings exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-026
Test Case Name	PATCH /api/bookings/{id}/cancel: successfully cancels and sends email when email header present
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Successfully cancels and sends email when email header present
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-027
Test Case Name	PATCH /api/bookings/{id}/cancel: cancels without sending email when email header is omitted
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Cancels without sending email when email header is omitted
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-028
Test Case Name	PATCH /api/bookings/{id}/cancel: returns 400 when cancellation window expired
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when cancellation window expired
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-029
Test Case Name	PATCH /api/bookings/{id}/cancel: returns 403 when trying to cancel another user's booking
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 403 when trying to cancel another user's booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-030
Test Case Name	PATCH /api/bookings/{id}/reschedule: successfully reschedules and sends email
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Successfully reschedules and sends email
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-031
Test Case Name	PATCH /api/bookings/{id}/reschedule: reschedules as admin without email
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Reschedules as admin without email
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-032
Test Case Name	PATCH /api/bookings/{id}/reschedule: returns 409 when target slot unavailable
Related Req.	REQ-BKG-05
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 409 when target slot unavailable
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-033
Test Case Name	PATCH /api/bookings/{id}/confirm: successfully confirms booking
Related Req.	REQ-BKG-02
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Successfully confirms booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-034
Test Case Name	PATCH /api/bookings/{id}/confirm: returns 400 when paymentId is null
Related Req.	REQ-BKG-02
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when paymentId is null
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-035
Test Case Name	GET /api/bookings/admin/all: returns paginated bookings with default params
Related Req.	REQ-BKG-04
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns paginated bookings with default params
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-036
Test Case Name	GET /api/bookings/admin/all: supports status filter, custom page, size and sort
Related Req.	REQ-BKG-04
Related Test File	controller/ BookingControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Supports status filter, custom page, size and sort
Actual Output	Matches expected
Status	**Pass**

GameStationControllerTest - 12 test(s)

Test Case ID	TC-BKG-UNIT-037
Test Case Name	GET /api/stations: returns list of all game stations
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns list of all game stations
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-038
Test Case Name	GET /api/stations/{id}: returns station when found
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns station when found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-039
Test Case Name	GET /api/stations/{id}: returns 404 when station not found
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 404 when station not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-040
Test Case Name	GET /api/stations/{id}/availability: returns hourly slot availability for a date
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns hourly slot availability for a date
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-041
Test Case Name	POST /api/stations: creates new station successfully
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Creates new station successfully
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-042
Test Case Name	POST /api/stations: returns 400 when stationCode is blank
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when stationCode is blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-043
Test Case Name	POST /api/stations: returns 400 when hourlyRate is zero or negative
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 when hourlyRate is zero or negative
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-044
Test Case Name	POST /api/stations: returns 409 when stationCode already exists
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 409 when stationCode already exists
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-045
Test Case Name	PUT /api/stations/{id}: updates station successfully
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Updates station successfully
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-046
Test Case Name	PUT /api/stations/{id}: returns 404 when station to update is not found
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 404 when station to update is not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-047
Test Case Name	DELETE /api/stations/{id}: soft-deletes station returning 204 No Content
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Soft-deletes station returning 204 No Content
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-048
Test Case Name	DELETE /api/stations/{id}: returns 404 when station not found
Related Req.	REQ-BKG-07
Related Test File	controller/ GameStationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 404 when station not found
Actual Output	Matches expected
Status	**Pass**

GlobalExceptionHandlerTest - 9 test(s)

Test Case ID	TC-BKG-UNIT-049
Test Case Name	handleBookingException: returns status and error response from exception
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns status and error response from exception
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-050
Test Case Name	handleValidation: formats field error message
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Formats field error message
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-051
Test Case Name	handleValidation: handles binding result with no field errors
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Handles binding result with no field errors
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-052
Test Case Name	handleAccessDenied: returns 401 when anonymous or unauthenticated
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns 401 when anonymous or unauthenticated
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-053
Test Case Name	handleAccessDenied: returns 403 when authenticated user lacks permission
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns 403 when authenticated user lacks permission
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-054
Test Case Name	handleUnexpected: returns 500 INTERNAL_SERVER_ERROR
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns 500 INTERNAL_SERVER_ERROR
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-055
Test Case Name	handleMissingRequestHeader: returns 401 when X-User-Id or X-User-Role missing
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns 401 when X-User-Id or X-User-Role missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-056
Test Case Name	handleMissingRequestHeader: returns 400 when other header is missing
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns 400 when other header is missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-057
Test Case Name	handleTypeMismatch: returns custom message for UUID vs generic argument
Related Req.	REQ-BKG-11
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns custom message for UUID vs generic argument
Actual Output	Matches expected
Status	**Pass**

GameServiceTest - 29 test(s)

Test Case ID	TC-BKG-UNIT-058
Test Case Name	getAllGames: returns all games sorted by createdAt desc when no filters provided
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns all games sorted by createdAt desc when no filters provided
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-059
Test Case Name	getAllGames: filters by search query across title, genre, and description
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Filters by search query across title, genre, and description
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-060
Test Case Name	getAllGames: filters by genre, skipping when genre is ALL or blank
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Filters by genre, skipping when genre is ALL or blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-061
Test Case Name	getAllGames: filters by platform, skipping when platform is ALL or blank
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Filters by platform, skipping when platform is ALL or blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-062
Test Case Name	getAllGames: filters by active and featured flags
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Filters by active and featured flags
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-063
Test Case Name	getGameById: returns GameResponse when game exists
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns GameResponse when game exists
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-064
Test Case Name	getGameById: throws NOT_FOUND when game does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when game does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-065
Test Case Name	createGame: successfully creates game when title is unique
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully creates game when title is unique
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-066
Test Case Name	createGame: applies default values when optional fields are null or empty
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Applies default values when optional fields are null or empty
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-067
Test Case Name	createGame: throws CONFLICT when game title already exists
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws CONFLICT when game title already exists
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-068
Test Case Name	updateGame: successfully updates existing game when title unchanged
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully updates existing game when title unchanged
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-069
Test Case Name	updateGame: successfully updates when title changed to a new unique title
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully updates when title changed to a new unique title
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-070
Test Case Name	updateGame: throws CONFLICT when new title is already taken by another game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws CONFLICT when new title is already taken by another game
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-071
Test Case Name	updateGame: throws NOT_FOUND when game does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when game does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-072
Test Case Name	deleteGame: successfully deletes existing game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully deletes existing game
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-073
Test Case Name	deleteGame: throws NOT_FOUND when game does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when game does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-074
Test Case Name	toggleActive: toggles active status of game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Toggles active status of game
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-075
Test Case Name	toggleActive: throws NOT_FOUND when game does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when game does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-076
Test Case Name	getGameStats: returns aggregate statistics with top genre
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns aggregate statistics with top genre
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-077
Test Case Name	getGameStats: defaults top genre to ACTION when genres list is empty or null
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Defaults top genre to ACTION when genres list is empty or null
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-078
Test Case Name	getGamesForStation: returns games installed on station
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns games installed on station
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-079
Test Case Name	getGamesForStation: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-080
Test Case Name	assignGamesToStation: assigns new games and unassigns removed games
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Assigns new games and unassigns removed games
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-081
Test Case Name	assignGamesToStation: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-082
Test Case Name	removeGameFromStation: removes station from game if contained
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Removes station from game if contained
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-083
Test Case Name	removeGameFromStation: does nothing if station not attached to game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Does nothing if station not attached to game
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-084
Test Case Name	removeGameFromStation: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-085
Test Case Name	getStationsForGame: returns list of stations assigned to game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns list of stations assigned to game
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-086
Test Case Name	updateStationsForGame: updates stations for game
Related Req.	REQ-BKG-06
Related Test File	service/ GameServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Updates stations for game
Actual Output	Matches expected
Status	**Pass**

GameStationServiceTest - 14 test(s)

Test Case ID	TC-BKG-UNIT-087
Test Case Name	createStation: successfully creates station when code is unique
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully creates station when code is unique
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-088
Test Case Name	createStation: throws CONFLICT when station code already exists
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws CONFLICT when station code already exists
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-089
Test Case Name	updateStation: updates station successfully without code change
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Updates station successfully without code change
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-090
Test Case Name	updateStation: updates station with new unique code
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Updates station with new unique code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-091
Test Case Name	updateStation: throws CONFLICT when new code is taken by another station
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws CONFLICT when new code is taken by another station
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-092
Test Case Name	updateStation: allows updating when findByStationCode returns self
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	updateStation; allows updating when findByStationCode
Expected Output	Returns self
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-093
Test Case Name	updateStation: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-094
Test Case Name	deleteStation: soft-deletes station by setting active to false
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Soft-deletes station by setting active to false
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-095
Test Case Name	deleteStation: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-096
Test Case Name	getAllStations: returns all mapped stations
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns all mapped stations
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-097
Test Case Name	getStationById: returns station response when found
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns station response when found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-098
Test Case Name	getStationById: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-099
Test Case Name	getStationAvailability: generates 13 hourly slots and correctly checks conflicts
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates 13 hourly slots and correctly checks conflicts
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-100
Test Case Name	getStationAvailability: throws NOT_FOUND when station does not exist
Related Req.	REQ-BKG-07
Related Test File	service/ GameStationServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws NOT_FOUND when station does not exist
Actual Output	Matches expected
Status	**Pass**

ReviewServiceTest - 3 test(s)

Test Case ID	**TC-BKG-UNIT-101**
Test Case Name	addReview - successful review creation for confirmed booking
Related Req.	REQ-BKG-08
Related Test File	service/ ReviewServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successful review creation for confirmed booking
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	**TC-BKG-UNIT-102**
Test Case Name	addReview - throws 403 Forbidden when user is not booking owner
Related Req.	REQ-BKG-08
Related Test File	service/ ReviewServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 403 Forbidden when user is not booking owner
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	**TC-BKG-UNIT-103**
Test Case Name	addReview - throws 409 Conflict when review already exists
Related Req.	REQ-BKG-08
Related Test File	service/ ReviewServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 Conflict when review already exists
Actual Output	Added after last recorded run
Status	**Not Run**

BookingServiceTest - 46 test(s)

Test Case ID	**TC-BKG-UNIT-104**
Test Case Name	createBooking: succeeds given free slot for walk-in booking
Related Req.	REQ-BKG-01
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Succeeds given free slot for walk-in booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-105
Test Case Name	createBooking: saves the booking with the selected game
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Saves the booking with the selected game
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-106
Test Case Name	createBooking: rejects overlapping booking
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Rejects overlapping booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-107
Test Case Name	createBooking: succeeds given adjacent slot
Related Req.	REQ-BKG-01
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Succeeds given adjacent slot
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-108
Test Case Name	createBooking: ignores cancelled bookings
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Ignores cancelled bookings
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-109
Test Case Name	createBooking: throws 404 when station does not exist
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 when station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-110
Test Case Name	createBooking: throws 409 CONFLICT when station is inactive
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when station is inactive
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-111
Test Case Name	createBooking: rejects end time before or equal to start time
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Rejects end time before or equal to start time
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-112
Test Case Name	createBooking: throws 400 GAME_ID_REQUIRED when gameId is missing
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 GAME_ID_REQUIRED when gameId is missing
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-113
Test Case Name	createBooking: throws 404 GAME_NOT_FOUND when game does not exist
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 GAME_NOT_FOUND when game does not exist
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-114
Test Case Name	createBooking: throws 409 GAME_INACTIVE when game is inactive
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 GAME_INACTIVE when game is inactive
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-115
Test Case Name	createBooking: throws 400 GAME_NOT_ON_STATION when game is not installed on the station
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 GAME_NOT_ON_STATION when game is not installed on the station
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-116
Test Case Name	createBooking: ONLINE happy path charges fee, confirms booking, and dispatches confirmation email
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	ONLINE happy path charges fee, confirms booking, and dispatches confirmation email
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-117
Test Case Name	createBooking: ONLINE booking skips email if user has no email
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	ONLINE booking skips email if user has no email
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-118
Test Case Name	createBooking: ONLINE booking succeeds even if notification client throws exception
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	ONLINE booking succeeds even if notification client throws exception
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-119
Test Case Name	createBooking: ONLINE marks booking CANCELLED and rethrows when PaymentClient throws BookingException
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	ONLINE marks booking CANCELLED and rethrows when PaymentClient throws BookingException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-120
Test Case Name	createBooking: ONLINE marks booking CANCELLED when PaymentClient returns null result
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	createBooking; ONLINE marks booking CANCELLED when PaymentClient
Expected Output	Returns null result
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-121
Test Case Name	getBookingsForUser: returns list of user bookings
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns list of user bookings
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-122
Test Case Name	cancelBooking: customer cancels own booking and triggers loyalty reversal
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Customer cancels own booking and triggers loyalty reversal
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-123
Test Case Name	cancelBooking: admin can cancel another user's booking
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Admin can cancel another user's booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-124
Test Case Name	cancelBooking: throws 404 when booking does not exist
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 when booking does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-125
Test Case Name	cancelBooking: throws 403 when non-admin tries to cancel someone else's booking
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 403 when non-admin tries to cancel someone else's booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-126
Test Case Name	cancelBooking: throws 409 CONFLICT when booking is already cancelled
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when booking is already cancelled
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-127
Test Case Name	cancelBooking: throws 400 BAD_REQUEST when within 2 hours of scheduled start
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 BAD_REQUEST when within 2 hours of scheduled start
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-128
Test Case Name	rescheduleBooking: succeeds for confirmed booking with valid new slot
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Succeeds for confirmed booking with valid new slot
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-129
Test Case Name	rescheduleBooking: falls back to the booking's existing game when request has no gameId
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Falls back to the booking's existing game when request has no gameId
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-130
Test Case Name	rescheduleBooking: throws 400 GAME_NOT_ON_STATION when game is not on the new station
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 GAME_NOT_ON_STATION when game is not on the new station
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-BKG-UNIT-131
Test Case Name	rescheduleBooking: admin can reschedule another user's booking
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Admin can reschedule another user's booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-132
Test Case Name	rescheduleBooking: throws 404 when booking does not exist
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 when booking does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-133
Test Case Name	rescheduleBooking: throws 403 when non-admin reschedules another user's booking
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 403 when non-admin reschedules another user's booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-134
Test Case Name	rescheduleBooking: throws 409 CONFLICT when booking status is not CONFIRMED
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when booking status is not CONFIRMED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-135
Test Case Name	rescheduleBooking: throws 400 BAD_REQUEST when within 2 hours of current start time
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 BAD_REQUEST when within 2 hours of current start time
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-136
Test Case Name	rescheduleBooking: throws 400 BAD_REQUEST when new endTime is before or equal to startTime
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 400 BAD_REQUEST when new endTime is before or equal to startTime
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-137
Test Case Name	rescheduleBooking: throws 404 when target station does not exist
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 when target station does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-138
Test Case Name	rescheduleBooking: throws 409 CONFLICT when target station is inactive
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when target station is inactive
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-139
Test Case Name	rescheduleBooking: throws 409 CONFLICT when overlapping with another booking
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when overlapping with another booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-140
Test Case Name	rescheduleBooking: succeeds without conflict when overlap is with self
Related Req.	REQ-BKG-05
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Succeeds without conflict when overlap is with self
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-141
Test Case Name	confirmBooking: rejects null paymentId
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Rejects null paymentId
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-142
Test Case Name	confirmBooking: throws 404 when booking does not exist
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 404 when booking does not exist
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-143
Test Case Name	confirmBooking: throws 409 CONFLICT when booking status is not PENDING
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws 409 CONFLICT when booking status is not PENDING
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-144
Test Case Name	confirmBooking: successfully confirms pending booking and sets paymentId
Related Req.	REQ-BKG-02
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully confirms pending booking and sets paymentId
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-145
Test Case Name	getAllBookings: delegates to findByStatus when status provided
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Delegates to findByStatus when status provided
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-146
Test Case Name	getAllBookings: delegates to findAll when status is null
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Delegates to findAll when status is null
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-147
Test Case Name	getAdminBookings: returns mapped admin booking rows and builds specification
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns mapped admin booking rows and builds specification
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-148
Test Case Name	getBookingStats: calculates aggregate statistics
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Calculates aggregate statistics
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-UNIT-149
Test Case Name	getBookingsForExport: returns list of admin booking rows for export
Related Req.	REQ-BKG-04
Related Test File	service/ service/BookingServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns list of admin booking rows for export
Actual Output	Matches expected
Status	**Pass**

BookingControllerIntegrationTest - 7 test(s)

Test Case ID	TC-BKG-INT-001
Test Case Name	Post bookings as customer should succeed
Related Req.	REQ-BKG-02
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	As customer
Expected Output	Succeed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-002
Test Case Name	Post walk in as customer should return 403
Related Req.	REQ-BKG-03
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	As customer
Expected Output	Return 403
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-003
Test Case Name	Post walk in as admin should succeed
Related Req.	REQ-BKG-03
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	As admin
Expected Output	Succeed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-004
Test Case Name	Cancel booking as different customer should return 403
Related Req.	REQ-BKG-05
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	As different customer
Expected Output	Return 403
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-005
Test Case Name	Cancel booking as admin should succeed even for someone elses booking
Related Req.	REQ-BKG-05
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	As admin even for someone elses booking
Expected Output	Succeed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-006
Test Case Name	Get my bookings should return only callers bookings
Related Req.	REQ-BKG-04
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Return only callers bookings
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-INT-007
Test Case Name	Any endpoint without identity headers should return 401
Related Req.	REQ-BKG-02
Related Test File	service/ controller/BookingControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Without identity headers
Expected Output	Return 401
Actual Output	Matches expected
Status	**Pass**

GatewayHeaderAuthFilterTest - 7 test(s)

Test Case ID	**TC-BKG-SEC-001**
Test Case Name	doFilter: proceeds without auth when X-User-Id header is missing
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Proceeds without auth when X-User-Id header is missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-BKG-SEC-002**
Test Case Name	doFilter: proceeds without auth when X-User-Role header is missing
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Proceeds without auth when X-User-Role header is missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-BKG-SEC-003**
Test Case Name	doFilter: proceeds without auth when headers are blank
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Proceeds without auth when headers are blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-BKG-SEC-004**
Test Case Name	doFilter: proceeds without auth when role is unsupported
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Proceeds without auth when role is unsupported
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-SEC-005
Test Case Name	doFilter: sets ROLE_CUSTOMER authentication for valid CUSTOMER header
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Sets ROLE_CUSTOMER authentication for valid CUSTOMER header
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-SEC-006
Test Case Name	doFilter: sets ROLE_ADMIN authentication for valid ADMIN header
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Sets ROLE_ADMIN authentication for valid ADMIN header
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-BKG-SEC-007
Test Case Name	doFilter: returns 401 Unauthorized when X-User-Id is not a valid UUID
Related Req.	REQ-BKG-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 Unauthorized when X-User-Id is not a valid UUID
Actual Output	Matches expected
Status	**Pass**

3.4 payment-service - 94 test cases

LoyaltyClientTest - 7 test(s)

Test Case ID	TC-PAY-UNIT-001
Test Case Name	reserve: returns ReserveResponse when successful
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Parses 200 OK loyalty reservation response
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-002
Test Case Name	reserve: throws LoyaltyRejectedException on 422 Unprocessable Entity
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws LoyaltyRejectedException on 422
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-003
Test Case Name	reserve: throws LoyaltyUnavailableException on 500 Internal Server Error
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws LoyaltyUnavailableException on 500
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-004
Test Case Name	confirm: posts to confirm endpoint
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST to /confirm with headers
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-005
Test Case Name	release: posts to release endpoint
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST to /release with reason
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-006
Test Case Name	award: posts to award endpoint
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches POST to /award with payment ID
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-007
Test Case Name	award: retries once when first attempt fails
Related Req.	REQ-PAY-02
Related Test File	client/ LoyaltyClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Retries once on transient failure
Actual Output	Matches expected
Status	**Pass**

InternalPaymentControllerTest - 2 test(s)

Test Case ID	TC-PAY-UNIT-008
Test Case Name	processInternalPayment: delegates to service and returns 201 Created
Related Req.	REQ-PAY-09
Related Test File	controller/ InternalPaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	processInternalPayment; delegates to service and
Expected Output	Internal endpoint returns 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-009
Test Case Name	processInternalPayment: throws IllegalArgumentException when customerUserId is null
Related Req.	REQ-PAY-09
Related Test File	controller/ InternalPaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	IllegalArgumentException when userId missing
Actual Output	Matches expected
Status	**Pass**

PaymentControllerTest - 11 test(s)

Test Case ID	TC-PAY-UNIT-010
Test Case Name	processPayment: extracts userId from authentication and returns 201 Created
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	processPayment; extracts userId from authentication and
Expected Output	Controller returns 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-011
Test Case Name	processCashPayment: delegates to service and returns 201 Created
Related Req.	REQ-PAY-03
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	processCashPayment; delegates to service and
Expected Output	Controller returns 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-012
Test Case Name	getPaymentReceipt: extracts userId and role and returns 200 OK
Related Req.	REQ-PAY-04
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	getPaymentReceipt; extracts userId and role and
Expected Output	Controller returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-013
Test Case Name	getMyPayments: delegates to service and returns list of user payments
Related Req.	REQ-PAY-05
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	getMyPayments; delegates to service and
Expected Output	Controller returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-014
Test Case Name	getAdminPayments: parses pagination, sorting, filters, and caps size at 100
Related Req.	REQ-PAY-06
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Pageable parsed and size capped at 100
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-015
Test Case Name	getPaymentStats: clamps trendDays between 1 and 365 and returns 200 OK
Related Req.	REQ-PAY-07
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	getPaymentStats; clamps trendDays between 1 and 365 and
Expected Output	trendDays clamped to 365
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-016
Test Case Name	exportPaymentsCsv: produces CSV formatted content with headers and values
Related Req.	REQ-PAY-08
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	text/csv content with correct headers
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-017
Test Case Name	handlePayHereNotification: returns 200 OK for valid query param notification
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 200 OK for valid query param notification
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-018
Test Case Name	handlePayHereNotification: parses payload from body map and returns 200 OK
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	handlePayHereNotification; parses payload from body map and
Expected Output	Returns 200 OK
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-019
Test Case Name	handlePayHereNotification: returns 400 Bad Request on IllegalArgumentException
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 400 Bad Request on IllegalArgumentException
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-020
Test Case Name	handlePayHereNotification: returns 500 on unexpected exception
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Returns 500 on unexpected exception
Actual Output	Added after last recorded run
Status	**Not Run**

PaymentMethodJacksonTest - 1 test(s)

Test Case ID	TC-PAY-UNIT-021
Test Case Name	Should deserialize legacy card and online values
Related Req.	REQ-PAY-01
Related Test File	entity/ PaymentMethodJacksonTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Jackson deserializes CARD, ONLINE, CASH
Actual Output	Matches expected
Status	**Pass**

GlobalExceptionHandlerTest - 6 test(s)

Test Case ID	TC-PAY-UNIT-022
Test Case Name	handleLoyaltyRejected: returns 422 Unprocessable Entity with error body
Related Req.	REQ-PAY-02
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps to 422 UNPROCESSABLE_ENTITY
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-023
Test Case Name	handleLoyaltyUnavailable: returns 503 Service Unavailable with error body
Related Req.	REQ-PAY-02
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps to 503 SERVICE_UNAVAILABLE
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-024
Test Case Name	handleIllegalArgument: returns 400 Bad Request with error body
Related Req.	REQ-PAY-01
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps to 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-025
Test Case Name	handleMethodArgumentNotValid: returns 400 Bad Request with field error message
Related Req.	REQ-PAY-01
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps validation error to 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-026
Test Case Name	handleAccessDenied: returns 403 Forbidden when user is authenticated
Related Req.	REQ-PAY-10
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps authenticated access denied to 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-027
Test Case Name	handleAccessDenied: returns 401 Unauthorized when anonymous
Related Req.	REQ-PAY-10
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Maps anonymous access denied to 401 UNAUTHORIZED
Actual Output	Matches expected
Status	**Pass**

MockPaymentGatewayTest - 2 test(s)

Test Case ID	TC-PAY-UNIT-028
Test Case Name	processPayment: returns true for regular amounts not ending in .13
Related Req.	REQ-PAY-01
Related Test File	gateway/ MockPaymentGatewayTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Mock gateway returns true for standard amounts
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-029
Test Case Name	processPayment: returns false when amount ends in .13
Related Req.	REQ-PAY-01
Related Test File	gateway/ MockPaymentGatewayTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Mock gateway returns false for .13 amounts
Actual Output	Matches expected
Status	**Pass**

PayHerePaymentGatewayTest - 3 test(s)

Test Case ID	TC-PAY-UNIT-030
Test Case Name	generatePayHereHash produces non-null uppercase MD5 hash
Related Req.	REQ-PAY-01
Related Test File	gateway/ PayHerePaymentGatewayTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	GeneratePayHereHash produces non-null uppercase MD5 hash
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-031
Test Case Name	verifyNotification accepts matching MD5 signature
Related Req.	REQ-PAY-01
Related Test File	gateway/ PayHerePaymentGatewayTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	VerifyNotification accepts matching MD5 signature
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-032
Test Case Name	processPayment returns true for positive amount
Related Req.	REQ-PAY-01
Related Test File	gateway/ PayHerePaymentGatewayTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	ProcessPayment returns true for positive amount
Actual Output	Added after last recorded run
Status	**Not Run**

PaymentServiceTest - 31 test(s)

Test Case ID	TC-PAY-UNIT-033
Test Case Name	Process payment should award once with net amount when successful without points
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With net amount when successful without points
Expected Output	SUCCESS status, loyalty awarded once
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-034
Test Case Name	Process payment should reserve confirm and award when successful with points
Related Req.	REQ-PAY-02
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When successful with points
Expected Output	Discount applied, loyalty reserved & confirmed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-035
Test Case Name	Process payment should save failed and release when gateway returns false
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When gateway returns false
Expected Output	FAILED status, loyalty points released
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-036
Test Case Name	Process payment should release when gateway throws
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When gateway throws
Expected Output	RuntimeException rethrown, loyalty released
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-037
Test Case Name	Process payment should reject when loyalty reservation rejected
Related Req.	REQ-PAY-02
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When loyalty reservation rejected
Expected Output	LoyaltyRejectedException propagated
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-038
Test Case Name	Process payment should honor existing idempotency key without loyalty calls
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Without loyalty calls
Expected Output	Existing payment returned without charging
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-039
Test Case Name	Process cash payment should require customer user id for booking
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	For booking
Expected Output	IllegalArgumentException when missing customerUserId
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-040
Test Case Name	Process cash payment should award customer not admin when booking success
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When booking success
Expected Output	Loyalty awarded to customer, recordedBy is admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-041
Test Case Name	Process payment should skip loyalty for tournament entry
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	For tournament entry
Expected Output	No loyalty reserve/award for tournaments
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-042
Test Case Name	processPayment: throws IllegalArgumentException when amount is zero or negative
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	IllegalArgumentException on zero amount
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-043
Test Case Name	processPayment: throws IllegalArgumentException when loyaltyPointsToRedeem is negative
Related Req.	REQ-PAY-02
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	IllegalArgumentException on negative points
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-044
Test Case Name	processPayment: releases loyalty points and throws when payableAmount is zero or negative
Related Req.	REQ-PAY-02
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Points released and IllegalArgumentException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-045
Test Case Name	processPayment: handles concurrent DataIntegrityViolationException by returning existing payment
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Idempotent recovery on concurrency collision
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-046
Test Case Name	processPayment: rethrows DataIntegrityViolationException if idempotencyKey lookup is empty
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Exception rethrown when key not found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-047
Test Case Name	processCashPayment: returns existing payment if idempotencyKey is already used
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Existing payment returned
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-048
Test Case Name	processCashPayment: throws IllegalArgumentException when amount is zero or negative
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	IllegalArgumentException on zero cash amount
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-049
Test Case Name	processCashPayment: allows tournament cash payment without customerUserId
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Tournament entry recorded for admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-050
Test Case Name	processCashPayment: recovers from DataIntegrityViolationException when existing found
Related Req.	REQ-PAY-03
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Recovers existing payment on collision
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-051
Test Case Name	getPaymentReceipt: throws IllegalArgumentException when payment does not exist
Related Req.	REQ-PAY-04
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	IllegalArgumentException when ID unknown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-052
Test Case Name	getPaymentReceipt: returns receipt when role is ADMIN
Related Req.	REQ-PAY-04
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Admin retrieves any customer receipt
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-053
Test Case Name	getPaymentReceipt: returns receipt when role is CUSTOMER and user owns payment
Related Req.	REQ-PAY-04
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Customer retrieves their own receipt
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-054
Test Case Name	getPaymentReceipt: throws IllegalArgumentException when role is CUSTOMER and user does not own payment
Related Req.	REQ-PAY-04
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	400 Bad Request / unauthorized on cross-user access
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-055
Test Case Name	getPaymentReceipt: throws IllegalArgumentException when role is unauthorized
Related Req.	REQ-PAY-04
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Unauthorized on invalid role
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-056
Test Case Name	getMyPayments: returns list of user payments
Related Req.	REQ-PAY-05
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Returns user payments list
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-057
Test Case Name	getAdminPayments: queries repository with specification and returns paged results
Related Req.	REQ-PAY-06
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	getAdminPayments; queries repository with specification and
Expected Output	Returns Page of PaymentResponse
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-058
Test Case Name	getPaymentStats: computes stats and revenue trend
Related Req.	REQ-PAY-07
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Computes today, month, totals, trend
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-UNIT-059
Test Case Name	handlePayHereNotification: marks payment SUCCESS and generates receipt on status_code 2
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Marks payment SUCCESS and generates receipt on status_code 2
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-060
Test Case Name	handlePayHereNotification: marks payment FAILED on status_code -1, -2, or -3
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Marks payment FAILED on status_code -1, -2, or -3
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-061
Test Case Name	handlePayHereNotification: throws IllegalArgumentException when signature is invalid or blank
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws IllegalArgumentException when signature is invalid or blank
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-062
Test Case Name	handlePayHereNotification: throws IllegalArgumentException when payment cannot be found
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws IllegalArgumentException when payment cannot be found
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-PAY-UNIT-063
Test Case Name	handlePayHereNotification: locates payment by referenceId or idempotencyKey
Related Req.	REQ-PAY-01
Related Test File	service/ PaymentServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Locates payment by referenceId or idempotencyKey
Actual Output	Added after last recorded run
Status	**Not Run**

InternalPaymentControllerIntegrationTest - 3 test(s)

Test Case ID	TC-PAY-INT-001
Test Case Name	POST /api/internal/payments: service successfully processes payment for customer
Related Req.	REQ-PAY-09
Related Test File	controller/ InternalPaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 201 Created, DB record saved
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-002
Test Case Name	POST /api/internal/payments: returns 400 Bad Request when customerUserId is missing
Related Req.	REQ-PAY-09
Related Test File	controller/ InternalPaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-003
Test Case Name	POST /api/internal/payments: returns 401 Unauthorized when X-Gateway-Secret is missing
Related Req.	REQ-PAY-10
Related Test File	controller/ InternalPaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

PaymentControllerIntegrationTest - 18 test(s)

Test Case ID	TC-PAY-INT-004
Test Case Name	POST /api/payments: successfully processes customer payment and saves to database
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 201, DB record with SUCCESS and GC receipt
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-005
Test Case Name	POST /api/payments: processes payment with loyalty points redemption
Related Req.	REQ-PAY-02
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 201, DB record with net amount and points used
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-006
Test Case Name	POST /api/payments: records FAILED status when gateway rejects amount ending in .13
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 201, DB record with FAILED status
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-007
Test Case Name	POST /api/payments: enforces idempotency by returning existing payment on duplicate request
Related Req.	REQ-PAY-01
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Duplicate request returns original payment, count = 1
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-008
Test Case Name	POST /api/payments: returns 401 Unauthorized when gateway secret is missing
Related Req.	REQ-PAY-10
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-INT-009
Test Case Name	POST /api/payments/cash: Admin successfully records cash payment for booking
Related Req.	REQ-PAY-03
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 201, DB record with recordedBy=adminId
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-010**
Test Case Name	POST /api/payments/cash: returns 400 Bad Request when customerUserId is missing for booking
Related Req.	REQ-PAY-03
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-011**
Test Case Name	POST /api/payments/cash: returns 403 Forbidden when called by CUSTOMER
Related Req.	REQ-PAY-10
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-012**
Test Case Name	GET /api/payments/{id}/receipt: customer successfully retrieves their own receipt
Related Req.	REQ-PAY-04
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK with receipt details
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-013**
Test Case Name	GET /api/payments/{id}/receipt: admin can retrieve any customer receipt
Related Req.	REQ-PAY-04
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK with receipt details
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-014**
Test Case Name	GET /api/payments/{id}/receipt: returns 400 when customer tries to access another user's receipt
Related Req.	REQ-PAY-04
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 400 Bad Request / unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-015**
Test Case Name	GET /api/payments/mine: returns list of payments belonging to the user
Related Req.	REQ-PAY-05
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK with 2 user payments
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-016**
Test Case Name	GET /api/payments/admin/all: Admin retrieves filtered paged payments
Related Req.	REQ-PAY-06
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK with filtered page
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-017**
Test Case Name	GET /api/payments/admin/all: returns 403 Forbidden when called by CUSTOMER
Related Req.	REQ-PAY-10
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-018**
Test Case Name	GET /api/payments/admin/stats: Admin retrieves dashboard stats
Related Req.	REQ-PAY-07
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK with dashboard stats
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-019**
Test Case Name	GET /api/payments/admin/stats: returns 403 Forbidden when called by CUSTOMER
Related Req.	REQ-PAY-10
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-020**
Test Case Name	GET /api/payments/admin/export.csv: Admin exports payment log to CSV
Related Req.	REQ-PAY-08
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 200 OK, text/csv MIME, attachment header
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-INT-021**
Test Case Name	GET /api/payments/admin/export.csv: returns 403 Forbidden when called by CUSTOMER
Related Req.	REQ-PAY-10
Related Test File	controller/ PaymentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	HTTP 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

GatewayHeaderAuthFilterTest - 10 test(s)

Test Case ID	**TC-PAY-SEC-001**
Test Case Name	doFilter: rejects request without X-Gateway-Secret header
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when secret missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-SEC-002**
Test Case Name	doFilter: rejects request with invalid X-Gateway-Secret header
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when secret invalid
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-SEC-003**
Test Case Name	doFilter: bypasses swagger and OpenAPI documentation endpoints
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P2
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Documentation endpoints bypass filter
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-SEC-004**
Test Case Name	doFilter: authenticates internal request as ROLE_SERVICE
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Internal path granted ROLE_SERVICE
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-SEC-005
Test Case Name	doFilter: rejects request when X-User-Id is missing or blank
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when userId missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-SEC-006
Test Case Name	doFilter: rejects request when X-User-Role is missing or blank
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when role missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-SEC-007
Test Case Name	doFilter: rejects request when X-User-Id is not a valid UUID
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when UUID malformed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-PAY-SEC-008
Test Case Name	doFilter: rejects request when X-User-Role is not CUSTOMER or ADMIN
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 when role unknown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-SEC-009**
Test Case Name	doFilter: authenticates valid CUSTOMER request
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Grants ROLE_CUSTOMER
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-PAY-SEC-010**
Test Case Name	doFilter: authenticates valid ADMIN request
Related Req.	REQ-PAY-10
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Grants ROLE_ADMIN
Actual Output	Matches expected
Status	**Pass**

3.5 tournament-service - 114 test cases

NotificationClientTest - 5 test(s)

Test Case ID	**TC-TRN-UNIT-001**
Test Case Name	sendTournamentRegistrationConfirmed sends POST request with service headers
Related Req.	REQ-TRN-05
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches registration email POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-002**
Test Case Name	sendTournamentCancelled sends POST request with service headers
Related Req.	REQ-TRN-04
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches cancellation email POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-003**
Test Case Name	sendTournamentResultsPublished sends POST request with service headers
Related Req.	REQ-TRN-09
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches results email POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-004**
Test Case Name	sendTournamentReminder sends POST request with service headers
Related Req.	REQ-TRN-11
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Dispatches reminder email POST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-005
Test Case Name	Notification methods should catch exception safely when remote service fails
Related Req.	REQ-TRN-11
Related Test File	client/ NotificationClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When remote service fails
Expected Output	Remote error logged without uncaught crash
Actual Output	Matches expected
Status	**Pass**

PaymentClientTest - 5 test(s)

Test Case ID	TC-TRN-UNIT-006
Test Case Name	chargeEntryFee should return paymentId when payment-service returns SUCCESS
Related Req.	REQ-TRN-05
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When payment-service returns SUCCESS
Expected Output	Returns payment UUID on SUCCESS
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-007
Test Case Name	chargeEntryFee should throw PaymentFailedException when payment-service returns FAILED status
Related Req.	REQ-TRN-05
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When payment-service returns FAILED status
Expected Output	Throws PaymentFailedException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-008
Test Case Name	chargeEntryFee should throw PaymentFailedException when payment-service returns 400 Bad Request
Related Req.	REQ-TRN-05
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When payment-service returns 400 Bad Request
Expected Output	Throws PaymentFailedException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-009
Test Case Name	chargeEntryFee should throw PaymentFailedException when payment-service returns 500 Server Error
Related Req.	REQ-TRN-05
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When payment-service returns 500 Server Error
Expected Output	Throws PaymentFailedException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-010
Test Case Name	chargeEntryFee should work without gateway secret header when secret is blank
Related Req.	REQ-TRN-05
Related Test File	client/ PaymentClientTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When secret is blank
Expected Output	Proceeds without secret header
Actual Output	Matches expected
Status	**Pass**

UserClientTest - 1 test(s)

Test Case ID	TC-TRN-UNIT-011
Test Case Name	getUserById should return UserSummaryDto when user-service returns 200 OK
Related Req.	REQ-TRN-05
Related Test File	client/ UserClientTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user-service returns 200 OK
Expected Output	Returns Optional of UserSummaryDto
Actual Output	Matches expected
Status	**Pass**

TournamentControllerTest - 12 test(s)

Test Case ID	TC-TRN-UNIT-012
Test Case Name	createTournament should call service and return 201 Created
Related Req.	REQ-TRN-01
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	HTTP 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-013
Test Case Name	getAllTournaments should return 200 OK with tournaments list
Related Req.	REQ-TRN-02
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With tournaments list
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-014
Test Case Name	getTournamentStats should return 200 OK with admin stats
Related Req.	REQ-TRN-10
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With admin stats
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-015
Test Case Name	getTournamentById should return 200 OK with tournament response
Related Req.	REQ-TRN-02
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With tournament response
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-016
Test Case Name	updateTournamentStatus should return 200 OK with updated tournament
Related Req.	REQ-TRN-03
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With updated tournament
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-017**
Test Case Name	cancelTournament should return 200 OK with cancelled tournament
Related Req.	REQ-TRN-04
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With cancelled tournament
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-018**
Test Case Name	registerForTournament should return 201 Created with registration response
Related Req.	REQ-TRN-05
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With registration response
Expected Output	HTTP 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-019**
Test Case Name	getRegistrations should return 200 OK with registrations list
Related Req.	REQ-TRN-06
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With registrations list
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-020**
Test Case Name	generateBracket should return 201 Created with generated bracket
Related Req.	REQ-TRN-07
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With generated bracket
Expected Output	HTTP 201 Created
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-021
Test Case Name	getBracket should return 200 OK with tournament bracket
Related Req.	REQ-TRN-08
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With tournament bracket
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-022
Test Case Name	recordMatchResult should return 200 OK with updated match
Related Req.	REQ-TRN-09
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With updated match
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-023
Test Case Name	sendPreTournamentReminders should trigger reminders and return 200 OK
Related Req.	REQ-TRN-11
Related Test File	controller/ TournamentControllerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	HTTP 200 OK
Actual Output	Matches expected
Status	**Pass**

GlobalExceptionHandlerTest - 9 test(s)

Test Case ID	TC-TRN-UNIT-024
Test Case Name	handleRegistrationClosed should return 409 CONFLICT
Related Req.	REQ-TRN-05
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-025
Test Case Name	handleNotFound should return 404 NOT_FOUND
Related Req.	REQ-TRN-02
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 404 Not Found
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-026
Test Case Name	handleDuplicate should return 409 CONFLICT
Related Req.	REQ-TRN-05
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-027
Test Case Name	handlePaymentFailed should return 402 PAYMENT_REQUIRED
Related Req.	REQ-TRN-05
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 402 Payment Required
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-028
Test Case Name	handleIllegalState should return 409 CONFLICT
Related Req.	REQ-TRN-07
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 409 Conflict
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-029
Test Case Name	handleIllegalArgument should return 400 BAD_REQUEST
Related Req.	REQ-TRN-09
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	HTTP 400 Bad Request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-030
Test Case Name	handleMethodArgumentNotValid should return 400 BAD_REQUEST with field error message
Related Req.	REQ-TRN-01
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With field error message
Expected Output	HTTP 400 Bad Request with field error
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-031
Test Case Name	handleAccessDenied should return 403 FORBIDDEN when user is authenticated
Related Req.	REQ-TRN-12
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user is authenticated
Expected Output	HTTP 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-032
Test Case Name	handleAccessDenied should return 401 UNAUTHORIZED when anonymous
Related Req.	REQ-TRN-12
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When anonymous
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

TournamentReminderSchedulerTest - 2 test(s)

Test Case ID	**TC-TRN-UNIT-033**
Test Case Name	schedulePreTournamentReminders should invoke tournamentService
Related Req.	REQ-TRN-11
Related Test File	scheduler/ TournamentReminderSchedulerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Invokes processUpcomingPreTournamentReminders
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-034**
Test Case Name	schedulePreTournamentReminders should catch exceptions gracefully without throwing
Related Req.	REQ-TRN-11
Related Test File	scheduler/ TournamentReminderSchedulerTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Service error handled gracefully without throwing
Actual Output	Matches expected
Status	**Pass**

BracketServiceImplTest - 14 test(s)

Test Case ID	**TC-TRN-UNIT-035**
Test Case Name	generateBracket should throw TournamentNotFoundException when tournament does not exist
Related Req.	REQ-TRN-07
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament does not exist
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-UNIT-036**
Test Case Name	generateBracket should throw IllegalStateException when bracket already exists
Related Req.	REQ-TRN-07
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When bracket already exists
Expected Output	IllegalStateException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-037
Test Case Name	generateBracket should throw IllegalStateException when fewer than 2 confirmed participants
Related Req.	REQ-TRN-07
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When fewer than 2 confirmed participants
Expected Output	IllegalStateException thrown (< 2 players)
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-038
Test Case Name	generateBracket should create paired matches for even number of participants
Related Req.	REQ-TRN-07
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	For even number of participants
Expected Output	Creates balanced Round 1 pairings
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-039
Test Case Name	generateBracket should create a bye match for odd number of participants
Related Req.	REQ-TRN-07
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	For odd number of participants
Expected Output	Creates bye match (auto-advancing participant 1)
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-040
Test Case Name	recordMatchResult should throw IllegalArgumentException when match does not exist
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When match does not exist
Expected Output	IllegalArgumentException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-041
Test Case Name	recordMatchResult should throw IllegalArgumentException when winner is neither participant
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When winner is neither participant
Expected Output	IllegalArgumentException on non-participant winner
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-042
Test Case Name	recordMatchResult should update match when current round is not yet fully completed
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When current round is not yet fully completed
Expected Output	Saves match winner without generating next round
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-043
Test Case Name	recordMatchResult should auto-generate next round matches when current round completes with multiple winners
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When current round completes with multiple winners
Expected Output	Generates round N+1 matches from winners
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-044
Test Case Name	recordMatchResult should determine champion, set tournament COMPLETED, and fan out results when 1 winner remains
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When 1 winner remains
Expected Output	Sets COMPLETED status, crowns champion, emails players
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-045
Test Case Name	recordMatchResult should fallback to winner ID string if userClient returns empty for champion
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	If userClient returns empty for champion
Expected Output	Formats champion ID when user client has no profile
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-046
Test Case Name	recordMatchResult should skip sending results email if participant email cannot be resolved
Related Req.	REQ-TRN-09
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	If participant email cannot be resolved
Expected Output	Skips email safely when recipient email is empty
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-047
Test Case Name	getBracket should return bracket response with matches sorted by round
Related Req.	REQ-TRN-08
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With matches sorted by round
Expected Output	Returns BracketResponse with sorted matches
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-048
Test Case Name	getBracket should throw IllegalStateException when no bracket generated yet
Related Req.	REQ-TRN-08
Related Test File	service/ BracketServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When no bracket generated yet
Expected Output	IllegalStateException thrown
Actual Output	Matches expected
Status	**Pass**

RegistrationServiceImplTest - 10 test(s)

Test Case ID	TC-TRN-UNIT-049
Test Case Name	registerForTournament two-arg method should delegate to four-arg method and complete registration
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Successfully registers via 2-arg method
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-050
Test Case Name	registerForTournament should throw TournamentNotFoundException when tournament does not exist
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament does not exist
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-051
Test Case Name	registerForTournament should throw RegistrationClosedException when tournament status is not UPCOMING
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament status is not UPCOMING
Expected Output	RegistrationClosedException on status
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-052
Test Case Name	registerForTournament should permit registration if previous registration was CANCELLED
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	If previous registration was CANCELLED
Expected Output	Re-registration allowed if previous cancelled
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-053
Test Case Name	registerForTournament should cancel registration and rethrow PaymentFailedException on payment error
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Registration marked CANCELLED, PaymentFailedException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-054
Test Case Name	registerForTournament should catch DataIntegrityViolationException and rethrow DuplicateRegistrationException
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	DuplicateRegistrationException on DB race condition
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-055
Test Case Name	registerForTournament should format Free entry fee when entryFee is null
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When entryFee is null
Expected Output	Formats entry fee as 'Free' in notification
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-056
Test Case Name	registerForTournament should skip confirmation email when resolved email is blank
Related Req.	REQ-TRN-05
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When resolved email is blank
Expected Output	Registration succeeds without email failure
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-057
Test Case Name	getRegistrationsForTournament should return registrations when tournament exists
Related Req.	REQ-TRN-06
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament exists
Expected Output	List of RegistrationResponse returned
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-058
Test Case Name	getRegistrationsForTournament should throw TournamentNotFoundException when tournament missing
Related Req.	REQ-TRN-06
Related Test File	service/ RegistrationServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament missing
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

TournamentServiceImplTest - 21 test(s)

Test Case ID	TC-TRN-UNIT-059
Test Case Name	createAndPublishTournament should save tournament with UPCOMING status and 0 participants
Related Req.	REQ-TRN-01
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With UPCOMING status and 0 participants
Expected Output	UPCOMING status, 0 participants, isRegistrationOpen=true
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-060
Test Case Name	getAllTournaments should return all tournaments mapped to responses
Related Req.	REQ-TRN-02
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	All tournaments mapped to response objects
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-061
Test Case Name	getTournamentById should return tournament when found
Related Req.	REQ-TRN-02
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When found
Expected Output	TournamentResponse returned
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-062
Test Case Name	getTournamentById should throw TournamentNotFoundException when not found
Related Req.	REQ-TRN-02
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When not found
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-063
Test Case Name	updateTournamentStatus should update and persist new status
Related Req.	REQ-TRN-03
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Status updated and persisted
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-064
Test Case Name	updateTournamentStatus should throw TournamentNotFoundException when not found
Related Req.	REQ-TRN-03
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When not found
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-065
Test Case Name	cancelTournament should cancel tournament and fan out notifications to confirmed and pending participants
Related Req.	REQ-TRN-04
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Status set to CANCELLED, emails dispatched to all participants
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-066
Test Case Name	cancelTournament should skip sending email if user email cannot be resolved
Related Req.	REQ-TRN-04
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	If user email cannot be resolved
Expected Output	No crash, email skipped when user email is absent
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-067
Test Case Name	cancelTournament should throw TournamentNotFoundException when tournament does not exist
Related Req.	REQ-TRN-04
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament does not exist
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-068
Test Case Name	isRegistrationOpen should return true when upcoming, not full, and before deadline
Related Req.	REQ-TRN-05
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When upcoming, not full, and before deadline
Expected Output	Returns true when upcoming, not full, before deadline
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-069
Test Case Name	isRegistrationOpen should return false when participant count reached max
Related Req.	REQ-TRN-05
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When participant count reached max
Expected Output	Returns false when max participants reached
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-070
Test Case Name	isRegistrationOpen should return false when deadline passed
Related Req.	REQ-TRN-05
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When deadline passed
Expected Output	Returns false when deadline is in the past
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-071
Test Case Name	isRegistrationOpen should return false when status is not UPCOMING
Related Req.	REQ-TRN-05
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When status is not UPCOMING
Expected Output	Returns false when status is ONGOING/COMPLETED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-072
Test Case Name	isRegistrationOpen should throw TournamentNotFoundException when not found
Related Req.	REQ-TRN-05
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When not found
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-073
Test Case Name	processUpcomingPreTournamentReminders should trigger reminders for upcoming tournaments starting in <= 2h
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	For upcoming tournaments starting in <= 2h
Expected Output	Reminders sent for tournaments starting in <= 2 hours
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-074
Test Case Name	processUpcomingPreTournamentReminders should ignore tournaments that are not UPCOMING
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Non-UPCOMING tournaments skipped
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-075
Test Case Name	sendPreTournamentReminders should mark reminderSent when no confirmed registrations
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	When no confirmed registrations
Expected Output	reminderSent marked true without emails
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-076
Test Case Name	sendPreTournamentReminders should send reminder with opponent info and bye handling
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	With opponent info and bye handling
Expected Output	Opponent name and byes correctly computed in reminder
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-077
Test Case Name	sendPreTournamentReminders should resolve opponent name via userClient if userName missing
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	If userName missing
Expected Output	UserClient queried for opponent full name
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-078
Test Case Name	sendPreTournamentReminders should throw TournamentNotFoundException when tournament does not exist
Related Req.	REQ-TRN-11
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When tournament does not exist
Expected Output	TournamentNotFoundException thrown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-UNIT-079
Test Case Name	getTournamentStats should return aggregated tournament counts and participant sum
Related Req.	REQ-TRN-10
Related Test File	service/ TournamentServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Aggregated counts and participant sum computed
Actual Output	Matches expected
Status	**Pass**

TournamentControllerIntegrationTest - 26 test(s)

Test Case ID	TC-TRN-INT-001
Test Case Name	POST /api/tournaments - should create tournament when caller has ROLE_ADMIN
Related Req.	REQ-TRN-01
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When caller has ROLE_ADMIN
Expected Output	POST /api/tournaments as ADMIN: 201 CREATED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-002
Test Case Name	POST /api/tournaments - should return 403 Forbidden when caller is CUSTOMER
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When caller is CUSTOMER
Expected Output	POST /api/tournaments as CUSTOMER: 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-003
Test Case Name	POST /api/tournaments - should return 400 Bad Request on validation failure
Related Req.	REQ-TRN-01
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/tournaments as ADMIN: 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-004
Test Case Name	GET /api/tournaments - should permit guest browsing and return tournament list
Related Req.	REQ-TRN-02
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/tournaments as GUEST: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-005
Test Case Name	GET /api/tournaments/admin/stats - should return stats for ADMIN
Related Req.	REQ-TRN-10
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	For ADMIN
Expected Output	GET /api/tournaments/admin/stats as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-006**
Test Case Name	GET /api/tournaments/admin/stats - should return 403 Forbidden for CUSTOMER
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	For CUSTOMER
Expected Output	GET /api/tournaments/admin/stats as CUSTOMER: 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-007**
Test Case Name	GET /api/tournaments/{id} - should return 200 OK when tournament exists
Related Req.	REQ-TRN-02
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When tournament exists
Expected Output	GET /api/tournaments/{id} as GUEST: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-008**
Test Case Name	GET /api/tournaments/{id} - should return 404 Not Found when ID does not exist
Related Req.	REQ-TRN-02
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When ID does not exist
Expected Output	GET /api/tournaments/{id} as GUEST: 404 NOT_FOUND
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-009**
Test Case Name	PATCH /api/tournaments/{id}/status - should update status when ADMIN
Related Req.	REQ-TRN-03
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When ADMIN
Expected Output	PATCH /api/tournaments/{id}/status as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-010
Test Case Name	PATCH /api/tournaments/{id}/status - should return 403 Forbidden when CUSTOMER
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When CUSTOMER
Expected Output	PATCH /api/tournaments/{id}/status as CUSTOMER: 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-011
Test Case Name	POST /api/tournaments/{id}/cancel - should cancel tournament and fan out notifications when ADMIN
Related Req.	REQ-TRN-04
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When ADMIN
Expected Output	POST /api/tournaments/{id}/cancel as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-012
Test Case Name	POST /api/tournaments/{id}/register - should register customer, charge payment, and increment count
Related Req.	REQ-TRN-05
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	POST /api/tournaments/{id}/register as CUSTOMER: 201 CREATED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-013
Test Case Name	POST /api/tournaments/{id}/register - should reject duplicate registration with 409 Conflict
Related Req.	REQ-TRN-05
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	With 409 Conflict
Expected Output	POST /api/tournaments/{id}/register as CUSTOMER: 409 CONFLICT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-014**
Test Case Name	POST /api/tournaments/{id}/register - should reject with 409 Conflict when tournament is full
Related Req.	REQ-TRN-05
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	With 409 Conflict when tournament is full
Expected Output	POST /api/tournaments/{id}/register as CUSTOMER: 409 CONFLICT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-015**
Test Case Name	POST /api/tournaments/{id}/register - should handle payment failure with 402 Payment Required
Related Req.	REQ-TRN-05
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	With 402 Payment Required
Expected Output	POST /api/tournaments/{id}/register as CUSTOMER: 402 PAYMENT_REQUIRED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-016**
Test Case Name	GET /api/tournaments/{id}/registrations - should return registrations for ADMIN
Related Req.	REQ-TRN-06
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	For ADMIN
Expected Output	GET /api/tournaments/{id}/registrations as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-017**
Test Case Name	GET /api/tournaments/{id}/registrations - should return 403 Forbidden for CUSTOMER
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	For CUSTOMER
Expected Output	GET /api/tournaments/{id}/registrations as CUSTOMER: 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-018**
Test Case Name	POST /api/tournaments/{id}/bracket/generate - should create bracket & matches in DB for ADMIN
Related Req.	REQ-TRN-07
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	For ADMIN
Expected Output	POST /api/tournaments/{id}/bracket/generate as ADMIN: 201 CREATED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-019**
Test Case Name	POST /api/tournaments/{id}/bracket/generate - should return 409 Conflict when < 2 participants
Related Req.	REQ-TRN-07
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When < 2 participants
Expected Output	POST /api/tournaments/{id}/bracket/generate as ADMIN: 409 CONFLICT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-020**
Test Case Name	GET /api/tournaments/{id}/bracket - should return bracket and matches
Related Req.	REQ-TRN-08
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	GET /api/tournaments/{id}/bracket as GUEST: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-TRN-INT-021**
Test Case Name	PATCH /api/tournaments/{id}/bracket/matches/{matchId}/result - should record winner and crown champion
Related Req.	REQ-TRN-09
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	PATCH /api/tournaments/{id}/bracket/matches/{matchId}/result as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-022
Test Case Name	POST /api/tournaments/{id}/notify-reminders - should trigger reminders when ADMIN
Related Req.	REQ-TRN-11
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When ADMIN
Expected Output	POST /api/tournaments/{id}/notify-reminders as ADMIN: 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-023
Test Case Name	Security: should return 401 Unauthorized when X-Gateway-Secret header is missing
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When X-Gateway-Secret header is missing
Expected Output	POST /api/tournaments as NONE: 401 UNAUTHORIZED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-024
Test Case Name	Security: should return 401 Unauthorized when X-Gateway-Secret is invalid
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When X-Gateway-Secret is invalid
Expected Output	POST /api/tournaments as NONE: 401 UNAUTHORIZED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-025
Test Case Name	Security: should return 401 Unauthorized when X-User-Id is malformed
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When X-User-Id is malformed
Expected Output	POST /api/tournaments as NONE: 401 UNAUTHORIZED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-INT-026
Test Case Name	Security: should return 401 Unauthorized when X-User-Role is unsupported
Related Req.	REQ-TRN-12
Related Test File	controller/ TournamentControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	When X-User-Role is unsupported
Expected Output	POST /api/tournaments as NONE: 401 UNAUTHORIZED
Actual Output	Matches expected
Status	**Pass**

GatewayHeaderAuthFilterTest - 9 test(s)

Test Case ID	TC-TRN-SEC-001
Test Case Name	Should bypass swagger, openapi, actuator, and error endpoints without secret check
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Filter chain proceeds without 401
Actual Output	Matches expected (5 variants)
Status	**Pass**

Test Case ID	TC-TRN-SEC-002
Test Case Name	Should reject requests missing X-Gateway-Secret with 401 UNAUTHORIZED
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-003
Test Case Name	Should reject requests with incorrect X-Gateway-Secret with 401 UNAUTHORIZED
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-004
Test Case Name	Should permit guest browsing for GET /api/tournaments without userId header
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Guest allowed for GET /api/tournaments
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-005
Test Case Name	Should reject requests with missing X-User-Id header
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-006
Test Case Name	Should reject requests with missing X-User-Role header
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-007
Test Case Name	Should reject requests with malformed UUID in X-User-Id
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	HTTP 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-008
Test Case Name	Should authenticate valid CUSTOMER and set ROLE_CUSTOMER authority
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	ROLE_CUSTOMER set in SecurityContext
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-TRN-SEC-009
Test Case Name	Should authenticate valid ADMIN and set ROLE_ADMIN authority
Related Req.	REQ-TRN-12
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	ROLE_ADMIN set in SecurityContext
Actual Output	Matches expected
Status	**Pass**

3.6 loyalty-service - 123 test cases

LoyaltyAdminControllerTest - 6 test(s)

Test Case ID	TC-LOY-UNIT-001
Test Case Name	Given rules requested, when getRules, then returns 200 OK and rules response
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Rules requested; when getRules
Expected Output	Returns 200 OK and rules response
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-002
Test Case Name	Given valid update request, when updateRules, then returns 200 OK and updated rules
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Valid update request; when updateRules
Expected Output	Returns 200 OK and updated rules
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-003
Test Case Name	Given valid adjust request, when adjustPoints, then returns 200 OK and adjustment response
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Valid adjust request; when adjustPoints
Expected Output	Returns 200 OK and adjustment response
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-004
Test Case Name	Given stats requested, when getLoyaltyStats, then returns 200 OK and admin stats
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Stats requested; when getLoyaltyStats
Expected Output	Returns 200 OK and admin stats
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-005
Test Case Name	Given accounts requested, when getAllAccounts, then returns 200 OK and paged accounts
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Accounts requested; when getAllAccounts
Expected Output	Returns 200 OK and paged accounts
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-006
Test Case Name	Given transactions requested, when getAllTransactions, then returns 200 OK and paged transactions
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyAdminControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Transactions requested; when getAllTransactions
Expected Output	Returns 200 OK and paged transactions
Actual Output	Matches expected
Status	**Pass**

LoyaltyControllerTest - 10 test(s)

Test Case ID	TC-LOY-UNIT-007
Test Case Name	Given customer accesses own balance, when getBalance, then returns 200 OK and account response
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer accesses own balance; when getBalance
Expected Output	Returns 200 OK and account response
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-008**
Test Case Name	Given admin accesses any customer balance, when getBalance, then returns 200 OK
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Admin accesses any customer balance; when getBalance
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-009**
Test Case Name	Given customer accesses another user balance, when getBalance, then throws ForbiddenOperationException
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer accesses another user balance; when getBalance
Expected Output	Throws ForbiddenOperationException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-010**
Test Case Name	Given caller is null, when getBalance, then throws ForbiddenOperationException
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Caller is null; when getBalance
Expected Output	Throws ForbiddenOperationException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-011**
Test Case Name	Given customer accesses own history, when getHistory, then returns 200 OK and paged history
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer accesses own history; when getHistory
Expected Output	Returns 200 OK and paged history
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-012
Test Case Name	Given admin accesses customer history, when getHistory, then returns 200 OK
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Admin accesses customer history; when getHistory
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-013
Test Case Name	Given customer accesses another user history, when getHistory, then throws ForbiddenOperationException
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer accesses another user history; when getHistory
Expected Output	Throws ForbiddenOperationException
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-014
Test Case Name	Given customer calculates own redemption, when calculateRedemption, then returns 200 OK
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer calculates own redemption; when calculateRedemption
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-015
Test Case Name	Given admin calculates redemption for customer, when calculateRedemption, then returns 200 OK
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Admin calculates redemption for customer; when calculateRedemption
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-016
Test Case Name	Given customer calculates redemption for another user, when calculateRedemption, then throws ForbiddenOperationException
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Customer calculates redemption for another user; when calculateRedemption
Expected Output	Throws ForbiddenOperationException
Actual Output	Matches expected
Status	**Pass**

LoyaltyInternalControllerTest - 1 test(s)

Test Case ID	TC-LOY-UNIT-017
Test Case Name	Internal endpoints should be mapped and reach controller
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyInternalControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	MockMvc request; service layer mocked
Expected Output	Be mapped and reach controller
Actual Output	Matches expected (5 variants)
Status	**Pass**

GlobalExceptionHandlerTest - 14 test(s)

Test Case ID	TC-LOY-UNIT-018
Test Case Name	Given MethodArgumentNotValidException, when handled, then returns 400 BAD_REQUEST with field errors
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	MethodArgumentNotValidException; when handled
Expected Output	Returns 400 BAD_REQUEST with field errors
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-019
Test Case Name	Given InvalidLoyaltyRequestException, when handled, then returns 422 UNPROCESSABLE_ENTITY
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	InvalidLoyaltyRequestException; when handled
Expected Output	Returns 422 UNPROCESSABLE_ENTITY
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-020
Test Case Name	Given InsufficientPointsException, when handled, then returns 422 UNPROCESSABLE_ENTITY
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	InsufficientPointsException; when handled
Expected Output	Returns 422 UNPROCESSABLE_ENTITY
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-021
Test Case Name	Given RedemptionRejectedException, when handled, then returns 422 with specific error code
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	RedemptionRejectedException; when handled
Expected Output	Returns 422 with specific error code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-022
Test Case Name	Given MissingRequestHeaderException, when handled, then returns 400 BAD_REQUEST
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	MissingRequestHeaderException; when handled
Expected Output	Returns 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-023
Test Case Name	Given LoyaltyAccountNotFoundException, when handled, then returns 404 NOT_FOUND
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	LoyaltyAccountNotFoundException; when handled
Expected Output	Returns 404 NOT_FOUND
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-024
Test Case Name	Given LoyaltyRulesNotFoundException, when handled, then returns 404 NOT_FOUND
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	LoyaltyRulesNotFoundException; when handled
Expected Output	Returns 404 NOT_FOUND
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-025
Test Case Name	Given ForbiddenOperationException, when handled, then returns 403 FORBIDDEN
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	ForbiddenOperationException; when handled
Expected Output	Returns 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-026
Test Case Name	Given AccessDeniedException, when handled, then returns 403 FORBIDDEN
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	AccessDeniedException; when handled
Expected Output	Returns 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-027
Test Case Name	Given HttpMessageNotReadableException, when handled, then returns 400 BAD_REQUEST
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	HttpMessageNotReadableException; when handled
Expected Output	Returns 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-028**
Test Case Name	Given MethodArgumentTypeMismatchException, when handled, then returns 400 BAD_REQUEST
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	MethodArgumentTypeMismatchException; when handled
Expected Output	Returns 400 BAD_REQUEST
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-029**
Test Case Name	Given HttpRequestMethodNotSupportedException, when handled, then returns 405 METHOD_NOT_ALLOWED
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	HttpRequestMethodNotSupportedException; when handled
Expected Output	Returns 405 METHOD_NOT_ALLOWED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-030**
Test Case Name	Given DataIntegrityViolationException, when handled, then returns 409 CONFLICT
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	DataIntegrityViolationException; when handled
Expected Output	Returns 409 CONFLICT
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-031**
Test Case Name	Given unexpected Exception, when handled, then returns 500 INTERNAL_SERVER_ERROR
Related Req.	REQ-LOY-08
Related Test File	exception/ GlobalExceptionHandlerTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Unexpected Exception; when handled
Expected Output	Returns 500 INTERNAL_SERVER_ERROR
Actual Output	Matches expected
Status	**Pass**

LoyaltyReservationCleanupSchedulerTest - 4 test(s)

Test Case ID	**TC-LOY-UNIT-032**
Test Case Name	Given expired reservation booking IDs exist, when releaseExpiredReservations, then calls releaseRedemption for each
Related Req.	REQ-LOY-06
Related Test File	scheduler/ LoyaltyReservationCleanupSchedulerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Expired reservation booking IDs exist; when releaseExpiredReservations
Expected Output	Calls releaseRedemption for each
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-033**
Test Case Name	Given no expired reservations, when releaseExpiredReservations, then does not call releaseRedemption
Related Req.	REQ-LOY-06
Related Test File	scheduler/ LoyaltyReservationCleanupSchedulerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	No expired reservations; when releaseExpiredReservations
Expected Output	Does not call releaseRedemption
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-034**
Test Case Name	Given lookup throws exception, when releaseExpiredReservations, then handles gracefully without throwing
Related Req.	REQ-LOY-06
Related Test File	scheduler/ LoyaltyReservationCleanupSchedulerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Lookup throws exception; when releaseExpiredReservations
Expected Output	Handles gracefully without throwing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-LOY-UNIT-035**
Test Case Name	Given single release fails, when releaseExpiredReservations, then continues releasing others
Related Req.	REQ-LOY-06
Related Test File	scheduler/ LoyaltyReservationCleanupSchedulerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	Single release fails; when releaseExpiredReservations
Expected Output	Continues releasing others
Actual Output	Matches expected
Status	**Pass**

LoyaltyServiceTest - 40 test(s)

Test Case ID	TC-LOY-UNIT-036
Test Case Name	Get balance should return zero when user has no account
Related Req.	REQ-LOY-01
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When user has no account
Expected Output	Return zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-037
Test Case Name	Get balance should show lkr worth of points
Related Req.	REQ-LOY-01
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Show lkr worth of points
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-038
Test Case Name	Get rules should throw when rules are not configured
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When rules are not configured
Expected Output	Throw
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-039
Test Case Name	Get history should page in the database
Related Req.	REQ-LOY-01
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Page in the database
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-040
Test Case Name	Award points should add points and write ledger row
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Add points and write ledger row
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-041
Test Case Name	Award points should use silver multiplier from rules not ahard coded value
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Not ahard coded value
Expected Output	Use silver multiplier from rules
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-042
Test Case Name	Award points should upgrade tier when lifetime points reach silver threshold
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When lifetime points reach silver threshold
Expected Output	Upgrade tier
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-043
Test Case Name	Award points should be idempotent when booking already awarded
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When booking already awarded
Expected Output	Be idempotent
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-044
Test Case Name	Award points should add nothing when booking was already cancelled
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When booking was already cancelled
Expected Output	Add nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-045
Test Case Name	Calculate redemption should return discount and payable when valid
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When valid
Expected Output	Return discount and payable
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-046
Test Case Name	Calculate redemption should explain why invalid when below minimum
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When below minimum
Expected Output	Explain why invalid
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-047
Test Case Name	Reserve redemption should hold points as pending
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Hold points as pending
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-048
Test Case Name	Reserve redemption should return existing reservation even though balance is now lower
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Even though balance is now lower
Expected Output	Return existing reservation
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-049
Test Case Name	Reserve redemption should reject when below minimum
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When below minimum
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-050
Test Case Name	Reserve redemption should reject when not amultiple of the step
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When not amultiple of the step
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-051
Test Case Name	Reserve redemption should reject when balance is too low
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When balance is too low
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-052
Test Case Name	Reserve redemption should reject when discount exceeds the cap
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When discount exceeds the cap
Expected Output	Reject
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-053
Test Case Name	Confirm redemption should move pending to confirmed
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Move pending to confirmed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-054
Test Case Name	Confirm redemption should return no action when nothing is pending
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When nothing is pending
Expected Output	Return no action
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-055
Test Case Name	Release redemption should return points and write audit row
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Return points and write audit row
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-056
Test Case Name	Release redemption should return no action when nothing is pending
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When nothing is pending
Expected Output	Return no action
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-057
Test Case Name	Reverse booking should restore redeemed points and remove earned points
Related Req.	REQ-LOY-02
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Restore redeemed points and remove earned points
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-058
Test Case Name	Reverse booking should release apending reservation
Related Req.	REQ-LOY-04
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Release apending reservation
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-059
Test Case Name	Reverse booking should only remove what is available when points were already spent
Related Req.	REQ-LOY-04
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When points were already spent
Expected Output	Only remove what is available
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-060
Test Case Name	Reverse booking should do nothing when called asecond time
Related Req.	REQ-LOY-04
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When called asecond time
Expected Output	Do nothing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-061
Test Case Name	Reverse booking should return zeros when nothing exists for the booking
Related Req.	REQ-LOY-04
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When nothing exists for the booking
Expected Output	Return zeros
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-062
Test Case Name	Adjust points should store the signed value
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Store the signed value
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-063
Test Case Name	Adjust points should reject abalance below zero
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Reject abalance below zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-064
Test Case Name	Adjust points should reject zero
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Reject zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-065
Test Case Name	Update rules should save new values and record the admin
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Save new values and record the admin
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-066
Test Case Name	Update rules should reject silver threshold not below gold
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Reject silver threshold not below gold
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-067
Test Case Name	Update rules should reject gold multiplier below silver
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Reject gold multiplier below silver
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-068
Test Case Name	Find expired reservation booking ids should return bookings with old pending reservations
Related Req.	REQ-LOY-01
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Return bookings with old pending reservations
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-069
Test Case Name	Get loyalty stats should aggregate accounts and transactions by tier
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Aggregate accounts and transactions by tier
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-070
Test Case Name	Get all accounts should return paged accounts with lkr discount calculated
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Return paged accounts with lkr discount calculated
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-071
Test Case Name	Get all transactions should return paged transactions sorted desc
Related Req.	REQ-LOY-05
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Return paged transactions sorted desc
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-072
Test Case Name	Calculate redemption should throw invalid loyalty request when booking total is zero or negative
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When booking total is zero or negative
Expected Output	Throw invalid loyalty request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-073
Test Case Name	Reserve redemption should throw invalid loyalty request when booking already reserved by another user
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When booking already reserved by another user
Expected Output	Throw invalid loyalty request
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-074
Test Case Name	Confirm redemption should return confirmed when reservation already confirmed
Related Req.	REQ-LOY-03
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When reservation already confirmed
Expected Output	Return confirmed
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-075
Test Case Name	Reverse booking should downgrade tier when reversal reduces lifetime points
Related Req.	REQ-LOY-04
Related Test File	service/ LoyaltyServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When reversal reduces lifetime points
Expected Output	Downgrade tier
Actual Output	Matches expected
Status	**Pass**

PointsCalculatorTest - 10 test(s)

Test Case ID	TC-LOY-UNIT-076
Test Case Name	Tier multiplier should read values from rules
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Read values from rules
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-077
Test Case Name	Calculate earned points should floor for bronze
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Floor for bronze
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-078
Test Case Name	Calculate earned points should apply silver multiplier
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Apply silver multiplier
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-079
Test Case Name	Calculate earned points should apply gold multiplier and floor again
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Apply gold multiplier and floor again
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-080
Test Case Name	Calculate earned points should return zero when below one point
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When below one point
Expected Output	Return zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-081
Test Case Name	Calculate earned points should return zero when amount is zero or null
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	When amount is zero or null
Expected Output	Return zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-082
Test Case Name	Calculate discount should multiply points by point value
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Multiply points by point value
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-083
Test Case Name	Calculate payable amount should subtract discount and never go below zero
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Subtract discount and never go below zero
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-084
Test Case Name	Max discount should be the configured share of the total
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Be the configured share of the total
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-UNIT-085
Test Case Name	Is multiple of step should accept only multiples
Related Req.	REQ-LOY-02
Related Test File	util/ PointsCalculatorTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Accept only multiples
Actual Output	Matches expected
Status	**Pass**

LoyaltyServiceAdminIntegrationTest - 9 test(s)

Test Case ID	TC-LOY-INT-001
Test Case Name	GET /api/loyalty/admin/rules - admin caller returns 200 OK and current rules
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/admin/rules; admin caller
Expected Output	Returns 200 OK and current rules
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-002
Test Case Name	GET /api/loyalty/admin/rules - customer caller returns 403 FORBIDDEN
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/admin/rules; customer caller
Expected Output	Returns 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-003
Test Case Name	PUT /api/loyalty/admin/rules - admin updates rules and persists new values
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin updates rules and persists new values
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-004
Test Case Name	PUT /api/loyalty/admin/rules - customer caller returns 403 FORBIDDEN
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	PUT /api/loyalty/admin/rules; customer caller
Expected Output	Returns 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-005
Test Case Name	POST /api/loyalty/admin/adjust - admin adjusts points positively, balance updates, ledger saved
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin adjusts points positively, balance updates, ledger saved
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-006
Test Case Name	POST /api/loyalty/admin/adjust - adjustment that makes balance negative returns 422
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/loyalty/admin/adjust; adjustment that makes balance negative
Expected Output	Returns 422
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-007
Test Case Name	GET /api/loyalty/admin/stats - returns aggregated loyalty stats
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns aggregated loyalty stats
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-008
Test Case Name	GET /api/loyalty/admin/accounts - admin lists all accounts with pagination
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin lists all accounts with pagination
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-009
Test Case Name	GET /api/loyalty/admin/transactions - admin lists all transactions with pagination
Related Req.	REQ-LOY-05
Related Test File	controller/ LoyaltyServiceAdminIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Admin lists all transactions with pagination
Actual Output	Matches expected
Status	**Pass**

LoyaltyServiceCustomerIntegrationTest - 9 test(s)

Test Case ID	TC-LOY-INT-010
Test Case Name	GET /api/loyalty/balance/{userId} - customer accessing own balance returns 200 OK
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/balance/{userId}; customer accessing own balance
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-011
Test Case Name	GET /api/loyalty/balance/{userId} - user with no prior account gets default 0 balance
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	User with no prior account gets default 0 balance
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-012
Test Case Name	GET /api/loyalty/balance/{userId} - customer accessing another user returns 403 FORBIDDEN
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/balance/{userId}; customer accessing another user
Expected Output	Returns 403 FORBIDDEN
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-013
Test Case Name	GET /api/loyalty/balance/{userId} - admin accessing customer balance returns 200 OK
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/balance/{userId}; admin accessing customer balance
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-014
Test Case Name	GET /api/loyalty/history/{userId} - returns paginated history of transactions
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns paginated history of transactions
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-015
Test Case Name	GET /api/loyalty/history/{userId} - customer accessing other user history returns 403
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	GET /api/loyalty/history/{userId}; customer accessing other user history
Expected Output	Returns 403
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-016
Test Case Name	POST /api/loyalty/calculate-redemption - returns calculated discount and payable amount
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns calculated discount and payable amount
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-017
Test Case Name	POST /api/loyalty/calculate-redemption - returns valid=false when points are below minimum
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns valid=false when points are below minimum
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-018
Test Case Name	POST /api/loyalty/calculate-redemption - returns 403 when customer previews for another user
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceCustomerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns 403 when customer previews for another user
Actual Output	Matches expected
Status	**Pass**

LoyaltyServiceInternalIntegrationTest - 9 test(s)

Test Case ID	TC-LOY-INT-019
Test Case Name	POST /api/internal/loyalty/award - awards points for booking payment and updates balance & tier
Related Req.	REQ-LOY-02
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Awards points for booking payment and updates balance & tier
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-020
Test Case Name	POST /api/internal/loyalty/award - is idempotent when called again for same booking
Related Req.	REQ-LOY-02
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Is idempotent when called again for same booking
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-021
Test Case Name	POST /api/internal/loyalty/redeem/reserve - holds points as PENDING and returns discount
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	POST /api/internal/loyalty/redeem/reserve; holds points as PENDING and
Expected Output	Returns discount
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-022
Test Case Name	POST /api/internal/loyalty/redeem/reserve - returns 422 when balance is insufficient
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns 422 when balance is insufficient
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-023
Test Case Name	POST /api/internal/loyalty/redeem/confirm - moves reservation from PENDING to CONFIRMED
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Moves reservation from PENDING to CONFIRMED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-024
Test Case Name	POST /api/internal/loyalty/redeem/release - returns reserved points and sets status to RELEASED
Related Req.	REQ-LOY-03
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns reserved points and sets status to RELEASED
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-025
Test Case Name	POST /api/internal/loyalty/reverse - reverses earned points and restores confirmed redeemed points
Related Req.	REQ-LOY-02
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Reverses earned points and restores confirmed redeemed points
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-026
Test Case Name	POST /api/internal/loyalty/award - returns 401 UNAUTHORIZED when gateway secret is missing
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns 401 UNAUTHORIZED when gateway secret is missing
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-INT-027
Test Case Name	POST /api/internal/loyalty/award - returns 403 FORBIDDEN when user header is sent to internal endpoint
Related Req.	REQ-LOY-01
Related Test File	controller/ LoyaltyServiceInternalIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Returns 403 FORBIDDEN when user header is sent to internal endpoint
Actual Output	Matches expected
Status	**Pass**

GatewayAuthenticationFilterTest - 11 test(s)

Test Case ID	TC-LOY-SEC-001
Test Case Name	Do filter should return 401 when secret is missing
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When secret is missing
Expected Output	Return 401
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-002
Test Case Name	Do filter should return 401 when secret is wrong
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When secret is wrong
Expected Output	Return 401
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-003
Test Case Name	Do filter should allow actuator without secret
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Allow actuator without secret
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-004
Test Case Name	Do filter should grant service role on internal path with secret only
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	On internal path with secret only
Expected Output	Grant service role
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-005
Test Case Name	Do filter should return 403 when internal path carries auser identity
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When internal path carries auser identity
Expected Output	Return 403
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-006
Test Case Name	Do filter should return 401 when identity headers are missing
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When identity headers are missing
Expected Output	Return 401
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-007
Test Case Name	Do filter should return 401 when user id is not auuid
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When user id is not auuid
Expected Output	Return 401
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-008
Test Case Name	Do filter should authenticate acustomer
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Authenticate acustomer
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-009
Test Case Name	Do filter should accept arole prefixed admin role
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Accept arole prefixed admin role
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-010
Test Case Name	Do filter should return 403 when auser token claims the service role
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	When auser token claims the service role
Expected Output	Return 403
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-LOY-SEC-011
Test Case Name	Do filter should clear the security context after the request
Related Req.	REQ-LOY-07
Related Test File	security/ GatewayAuthenticationFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Clear the security context after the request
Actual Output	Matches expected
Status	**Pass**

3.7 notification-service - 70 test cases

NotificationControllerTest - 12 test(s)

Test Case ID	**TC-NTF-UNIT-001**
Test Case Name	sendWelcomeEmail: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendWelcomeEmail; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-NTF-UNIT-002**
Test Case Name	sendPasswordResetEmail: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendPasswordResetEmail; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-NTF-UNIT-003**
Test Case Name	sendAccountReactivationEmail: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendAccountReactivationEmail; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	**TC-NTF-UNIT-004**
Test Case Name	sendBookingConfirmation: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendBookingConfirmation; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-005
Test Case Name	sendBookingRescheduled: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendBookingRescheduled; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-006
Test Case Name	sendBookingCancelled: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendBookingCancelled; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-007
Test Case Name	sendTournamentRegistrationConfirmation: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendTournamentRegistrationConfirmation; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-008
Test Case Name	sendTournamentCancelled: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendTournamentCancelled; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-009
Test Case Name	sendTournamentResults: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendTournamentResults; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-010
Test Case Name	sendTournamentReminder: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendTournamentReminder; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-011
Test Case Name	sendPasswordResetSms: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-04
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendPasswordResetSms; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-012
Test Case Name	sendAccountReactivationSms: delegates to notificationService and returns 200 OK
Related Req.	REQ-NTF-04
Related Test File	controller/ NotificationControllerTest
Type	Unit
Pri.	P2
Given (Preconditions / Input)	sendAccountReactivationSms; delegates to notificationService and
Expected Output	Returns 200 OK
Actual Output	Matches expected
Status	**Pass**

EmailSenderServiceTest - 2 test(s)

Test Case ID	TC-NTF-UNIT-013
Test Case Name	sendHtmlEmail: creates MimeMessage and sends email successfully
Related Req.	REQ-NTF-05
Related Test File	service/ EmailSenderServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates MimeMessage and sends email successfully
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-014
Test Case Name	sendHtmlEmail: wraps MessagingException into RuntimeException
Related Req.	REQ-NTF-05
Related Test File	service/ EmailSenderServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Wraps MessagingException into RuntimeException
Actual Output	Matches expected
Status	**Pass**

EmailTemplateServiceTest - 10 test(s)

Test Case ID	TC-NTF-UNIT-015
Test Case Name	createWelcomeEmail: generates welcome email with formatted recipient name
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates welcome email with formatted recipient name
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-016
Test Case Name	createPasswordResetEmail: generates password reset email with formatted code
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates password reset email with formatted code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-017
Test Case Name	createAccountReactivationEmail: generates account reactivation email with code
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates account reactivation email with code
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-018
Test Case Name	createBookingConfirmationEmail: generates confirmation email with booking details
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates confirmation email with booking details
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-019
Test Case Name	createBookingRescheduledEmail: generates rescheduled email with new details
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates rescheduled email with new details
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-020
Test Case Name	createBookingCancelledEmail: generates cancellation email with refund policy note
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates cancellation email with refund policy note
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-021
Test Case Name	createTournamentRegistrationEmail: generates tournament registration confirmation
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates tournament registration confirmation
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-022
Test Case Name	createTournamentCancelledEmail: generates tournament cancellation notice
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates tournament cancellation notice
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-023
Test Case Name	createTournamentResultsEmail: generates tournament results published notice
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates tournament results published notice
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-024
Test Case Name	createTournamentReminderEmail: generates tournament reminder email
Related Req.	REQ-NTF-05
Related Test File	service/ EmailTemplateServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Generates tournament reminder email
Actual Output	Matches expected
Status	**Pass**

NotificationRecordServiceTest - 3 test(s)

Test Case ID	TC-NTF-UNIT-025
Test Case Name	recordSuccess: saves notification entity with status SENT and timestamps
Related Req.	REQ-NTF-06
Related Test File	service/ NotificationRecordServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Saves notification entity with status SENT and timestamps
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-026
Test Case Name	recordFailure: saves notification entity with status FAILED and error message
Related Req.	REQ-NTF-06
Related Test File	service/ NotificationRecordServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Saves notification entity with status FAILED and error message
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-027
Test Case Name	save: skips saving when recipient is null or blank
Related Req.	REQ-NTF-06
Related Test File	service/ NotificationRecordServiceTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Skips saving when recipient is null or blank
Actual Output	Matches expected
Status	**Pass**

NotificationServiceTest - 16 test(s)

Test Case ID	TC-NTF-UNIT-028
Test Case Name	sendWelcomeEmail: creates template, sends email, and records success
Related Req.	REQ-NTF-01
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates template, sends email, and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-029
Test Case Name	sendWelcomeEmail: records failure when email sending throws exception
Related Req.	REQ-NTF-01
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Records failure when email sending throws exception
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-030
Test Case Name	sendPasswordResetEmail: creates template, sends email, and records success
Related Req.	REQ-NTF-01
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates template, sends email, and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-031
Test Case Name	sendAccountReactivationEmail: creates template, sends email, and records success
Related Req.	REQ-NTF-01
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates template, sends email, and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-032
Test Case Name	sendBookingConfirmationEmail: sends email and broadcasts booking event to socket
Related Req.	REQ-NTF-02
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends email and broadcasts booking event to socket
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-033
Test Case Name	sendBookingRescheduledEmail: sends email and broadcasts rescheduled event to socket
Related Req.	REQ-NTF-02
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends email and broadcasts rescheduled event to socket
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-034
Test Case Name	sendBookingCancelledEmail: sends email and broadcasts cancellation event to socket
Related Req.	REQ-NTF-02
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends email and broadcasts cancellation event to socket
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-035
Test Case Name	sendTournamentRegistrationEmail: sends email and broadcasts registration event
Related Req.	REQ-NTF-03
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends email and broadcasts registration event
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-036
Test Case Name	sendTournamentCancelledEmail: sends email and broadcasts tournament cancelled event
Related Req.	REQ-NTF-03
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends email and broadcasts tournament cancelled event
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-037
Test Case Name	sendTournamentCancelledEmail: includes targetIds when userId is present
Related Req.	REQ-NTF-03
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Includes targetIds when userId is present
Actual Output	Added after last recorded run
Status	**Not Run**

Test Case ID	TC-NTF-UNIT-038
Test Case Name	sendTournamentResultsEmail: creates template, sends email, and records success
Related Req.	REQ-NTF-03
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates template, sends email, and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-039
Test Case Name	sendTournamentReminderEmail: creates template, sends email, and records success
Related Req.	REQ-NTF-03
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Creates template, sends email, and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-040
Test Case Name	sendPasswordResetSms: sends formatted SMS with expiration and records success
Related Req.	REQ-NTF-04
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends formatted SMS with expiration and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-041
Test Case Name	sendPasswordResetSms: records failure when SMS transport fails
Related Req.	REQ-NTF-04
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Records failure when SMS transport fails
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-042
Test Case Name	sendAccountReactivationSms: sends formatted SMS and records success
Related Req.	REQ-NTF-04
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends formatted SMS and records success
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-043
Test Case Name	sendAccountReactivationSms: records failure when SMS transport fails
Related Req.	REQ-NTF-04
Related Test File	service/ NotificationServiceTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Records failure when SMS transport fails
Actual Output	Matches expected
Status	**Pass**

SocketNotificationBroadcasterTest - 4 test(s)

Test Case ID	TC-NTF-UNIT-044
Test Case Name	broadcast: sends asynchronous POST request with notification payload
Related Req.	REQ-NTF-07
Related Test File	service/ SocketNotificationBroadcasterTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Sends asynchronous POST request with notification payload
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-045
Test Case Name	broadcast: handles null optional fields by defaulting to fallback values
Related Req.	REQ-NTF-07
Related Test File	service/ SocketNotificationBroadcasterTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Handles null optional fields by defaulting to fallback values
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-046
Test Case Name	broadcast: catches and ignores RestClientException if socket server is offline
Related Req.	REQ-NTF-07
Related Test File	service/ SocketNotificationBroadcasterTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Catches and ignores RestClientException if socket server is offline
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-047
Test Case Name	broadcast: includes targetUserId and targetUserIds in payload when provided
Related Req.	REQ-NTF-07
Related Test File	service/ SocketNotificationBroadcasterTest
Type	Unit
Pri.	P3
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Includes targetUserId and targetUserIds in payload when provided
Actual Output	Added after last recorded run
Status	**Not Run**

TwilioSmsServiceImplTest - 4 test(s)

Test Case ID	TC-NTF-UNIT-048
Test Case Name	send: normalizes local number starting with 0 to E.164 and creates Twilio message
Related Req.	REQ-NTF-04
Related Test File	service/ TwilioSmsServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Normalizes local number starting with 0 to E.164 and creates Twilio message
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-049
Test Case Name	send: keeps E.164 number when already prefixed with +
Related Req.	REQ-NTF-04
Related Test File	service/ TwilioSmsServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Keeps E.164 number when already prefixed with +
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-050
Test Case Name	send: prefixes with default country code when number starts without 0 or +
Related Req.	REQ-NTF-04
Related Test File	service/ TwilioSmsServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Prefixes with default country code when number starts without 0 or +
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-UNIT-051
Test Case Name	send: throws RuntimeException when Twilio throws ApiException
Related Req.	REQ-NTF-04
Related Test File	service/ TwilioSmsServiceImplTest
Type	Unit
Pri.	P1
Given (Preconditions / Input)	Dependencies mocked (Mockito)
Expected Output	Throws RuntimeException when Twilio throws ApiException
Actual Output	Matches expected
Status	**Pass**

NotificationControllerIntegrationTest - 14 test(s)

Test Case ID	TC-NTF-INT-001
Test Case Name	Security: rejects requests missing X-Gateway-Secret header with 401 Unauthorized
Related Req.	REQ-NTF-08
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Rejects requests missing X-Gateway-Secret header with 401 Unauthorized
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-002
Test Case Name	Security: rejects requests with non-SERVICE role with 403 Forbidden
Related Req.	REQ-NTF-08
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Rejects requests with non-SERVICE role with 403 Forbidden
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-003
Test Case Name	POST /api/notifications/welcome: sends welcome email and persists record
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends welcome email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-004
Test Case Name	POST /api/notifications/password-reset: sends password reset email and persists record
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends password reset email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-005
Test Case Name	POST /api/notifications/account-reactivation: sends reactivation email and persists record
Related Req.	REQ-NTF-01
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends reactivation email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-006
Test Case Name	POST /api/notifications/booking-confirmed: sends confirmation email and persists record
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends confirmation email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-007
Test Case Name	POST /api/notifications/booking-rescheduled: sends rescheduled email and persists record
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends rescheduled email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-008
Test Case Name	POST /api/notifications/booking-cancelled: sends cancellation email and persists record
Related Req.	REQ-NTF-02
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends cancellation email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-009
Test Case Name	POST /api/notifications/tournament-registration-confirmed: sends email and persists record
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-010
Test Case Name	POST /api/notifications/tournament-cancelled: sends email and persists record
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-011
Test Case Name	POST /api/notifications/tournament-results-published: sends email and persists record
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-012
Test Case Name	POST /api/notifications/tournament-reminder: sends email and persists record
Related Req.	REQ-NTF-03
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends email and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-013
Test Case Name	POST /api/notifications/password-reset-sms: sends reset SMS and persists record
Related Req.	REQ-NTF-04
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends reset SMS and persists record
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-INT-014
Test Case Name	POST /api/notifications/account-reactivation-sms: sends reactivation SMS and persists record
Related Req.	REQ-NTF-04
Related Test File	controller/ NotificationControllerIntegrationTest
Type	Integration
Pri.	P1
Given (Preconditions / Input)	Dedicated test database; HTTP request via test client
Expected Output	Sends reactivation SMS and persists record
Actual Output	Matches expected
Status	**Pass**

GatewayHeaderAuthFilterTest - 5 test(s)

Test Case ID	TC-NTF-SEC-001
Test Case Name	doFilterInternal: returns 401 Unauthorized when gateway secret is missing or incorrect
Related Req.	REQ-NTF-08
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 Unauthorized when gateway secret is missing or incorrect
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-SEC-002
Test Case Name	doFilterInternal: returns 401 Unauthorized when userId or role header is missing or blank
Related Req.	REQ-NTF-08
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 Unauthorized when userId or role header is missing or blank
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-SEC-003
Test Case Name	doFilterInternal: returns 401 Unauthorized when userId is not a valid UUID
Related Req.	REQ-NTF-08
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 Unauthorized when userId is not a valid UUID
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-SEC-004
Test Case Name	doFilterInternal: returns 401 Unauthorized when user role is unknown
Related Req.	REQ-NTF-08
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Returns 401 Unauthorized when user role is unknown
Actual Output	Matches expected
Status	**Pass**

Test Case ID	TC-NTF-SEC-005
Test Case Name	doFilterInternal: sets security context and proceeds down chain when headers valid
Related Req.	REQ-NTF-08
Related Test File	config/ GatewayHeaderAuthFilterTest
Type	Security
Pri.	P1
Given (Preconditions / Input)	Filter invoked with mocked request/exchange
Expected Output	Sets security context and proceeds down chain when headers valid
Actual Output	Matches expected
Status	**Pass**

4. Open Items

- Not-run tests (24). booking-service: BookingServiceTest (7); booking-service: ReviewServiceTest (3); notification-service: NotificationServiceTest (1); notification-service: SocketNotificationBroadcasterTest (1); payment-service: PayHerePaymentGatewayTest (3); payment-service: PaymentControllerTest (4); payment-service: PaymentServiceTest (5). Execute these with mvn verify and update the Status column.
- Requirement mapping. For user, booking, loyalty and notification services, requirements without an FR number in the code are labelled "Feature-level". Align them with the final SRS before submission.
- End-to-end tests. Frontend Playwright E2E cases are listed in Section 6. The latest reported Chromium run was 10 passed, including 8 Gaming Castle tests and 2 default Playwright example tests; the two example tests are excluded from this register. The booking-spec filenames should be checked against the current frontend tests/ directory before final submission. E2E coverage should be expanded to include remaining critical flows.
- Detail level. Given and Expected fields for user, booking, loyalty and notification tests are derived from test names and display names. Review the P1 cases and add precise input values where the course template needs them.
- Coverage and performance. JaCoCo coverage (deliverable 6) and performance results (deliverable 9) are separate documents and are not repeated here.

5. Frontend End-to-End (E2E) Test Register

The following records cover the existing Gaming Castle Playwright browser tests. The backend unit, integration, and security records above have been preserved. E2E tests exercise user-visible flows through the Angular application; they are separate from the backend service tests.

5.1 E2E Execution Summary

Registered E2E cases	Passed (reported)	Failed (reported)	Not run
8	8	0	0

Execution note: the previously reported full Chromium run was 10/10 passed, but two tests were default Playwright examples targeting playwright.dev and are not included above. The exact filenames for the three booking tests were not available in the source document, so they are flagged for verification.

5.2 Registered E2E Test Cases

Test Case ID	TC-E2E-001
Test Case Name	Application root loads and unauthenticated user is routed to login
Related Req.	Confirm against approved SRS
Related Test File	tests/app-smoke.spec.ts
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Frontend running; user is not authenticated; open application root
Expected Output	Application loads and unauthenticated navigation leads to login
Actual Output	Reported passed in prior Playwright run
Status	Pass*

Test Case ID	TC-E2E-002
Test Case Name	Login page displays heading, form controls, and submit button
Related Req.	Confirm against approved SRS
Related Test File	tests/app-smoke.spec.ts
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Frontend running; navigate to login page
Expected Output	Login heading, email/password fields, and submit button are visible
Actual Output	Reported passed in prior Playwright run
Status	Pass*

Test Case ID	TC-E2E-003
Test Case Name	Customer can log in with valid credentials
Related Req.	Confirm against approved SRS
Related Test File	tests/auth.spec.ts
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Dedicated customer test account configured through E2E environment variables; backend running
Expected Output	Customer reaches dashboard and customer session values are stored
Actual Output	Reported passed; successful-login test was also run individually
Status	Pass*

Test Case ID	TC-E2E-004
Test Case Name	Login rejects invalid credentials
Related Req.	Confirm against approved SRS
Related Test File	tests/auth.spec.ts
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Frontend/backend running; submit invalid test credentials
Expected Output	Error notification appears and user remains on login page
Actual Output	Reported passed in prior Playwright run
Status	Pass*

Test Case ID	TC-E2E-005
Test Case Name	Unauthenticated access to protected dashboard redirects to login
Related Req.	Confirm against approved SRS
Related Test File	tests/auth.spec.ts
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Clear authentication state; navigate directly to /dashboard
Expected Output	User is redirected to login and login form is visible
Actual Output	Reported passed in prior Playwright run
Status	Pass*

Test Case ID	TC-E2E-006
Test Case Name	Booking portal loads station catalogue and live summary
Related Req.	Confirm against approved SRS
Related Test File	Frontend tests/ booking spec — exact filename to verify
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Frontend/backend running; use a suitable customer session if required
Expected Output	Booking portal and station catalogue are displayed
Actual Output	Reported passed in prior full Playwright run; exact filename not verified
Status	Pass*

Test Case ID	TC-E2E-007
Test Case Name	Selecting station, future slot, and game updates pricing
Related Req.	Confirm against approved SRS
Related Test File	Frontend tests/ booking spec — exact filename to verify
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Booking portal available; select a station, future time slot, and game
Expected Output	Selected booking details are shown and price updates accordingly
Actual Output	Reported passed in prior full Playwright run; exact filename not verified
Status	Pass*

Test Case ID	TC-E2E-008
Test Case Name	Checkout flow displays payment modal and booking receipt
Related Req.	Confirm against approved SRS
Related Test File	Frontend tests/ booking spec — exact filename to verify
Type	E2E
Pri.	P1
Given (Preconditions / Input)	Booking checkout test setup and required backend services available
Expected Output	Checkout/payment modal completes and a booking receipt is shown
Actual Output	Reported passed in prior full Playwright run; backend persistence not independently confirmed
Status	Pass*

*Pass status reflects previously reported Playwright results in the conversation, not a new test run during document editing. Before submission, verify the three booking test filenames and map each case to the exact requirement IDs in the approved SRS. A checkout receipt assertion alone does not prove that booking/payment records were persisted in the backend.

5.3 Combined Register Notes

The backend register's original counts and individual records are retained. Its reported summary remains 775 registered tests, 751 passed, 0 failures recorded, and 24 not run. The E2E summary above is separate and must not be added to the backend totals until the final execution records are verified.